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Family-wise error control for correlated pathway tests in GWAS: CATFISH with Westfall–Young calibration

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Abstract

Motivation: Pathway analysis of genome-wide association study (GWAS) summary statistics must contend with two distinct sources of dependence. Genes within a pathway are correlated through linkage disequilibrium (LD), and pathways are correlated with one another through shared gene membership. Most pipelines address only the first, calibrating each pathway against its own LD-aware null and then applying a Bonferroni or Benjamini–Hochberg correction that assumes the resulting pathway p-values are independent. They are not, and the correction is consequently miscalibrated in a direction that depends on how strongly the pathway family overlaps.

Results: We extend CATFISH, a correlation-aware omnibus pathway framework combining ACAT, Fisher, adaptive soft TFisher, minP and Stouffer statistics, with a Westfall–Young max-statistic calibration that controls the family-wise error rate (FWER) exactly in the presence of arbitrary between-pathway dependence. A single genome-wide correlated null draw is propagated through every pathway in the family, the minimum omnibus p-value is recorded, and its distribution defines the family-wise threshold. The entire statistical pipeline is reimplemented in log space, which is necessary rather than cosmetic: in a GWAS of $n \approx 1.3$ million individuals the omnibus statistic underflows double precision for the majority of pathways, and the resulting ties silently destroy the ranking. Across simulated null families spanning three LD regimes and three pathway sizes, independence-based calibration was anti-conservative while MVN and Westfall–Young calibration held type I error near nominal; the calibrations remained stable when up to four of the five component tests were deliberately miscalibrated. Applied to a maize grain-nitrogen GWAS, the Westfall–Young ordering recovered L-homoserine biosynthesis as the leading pathway and placed a known aspartate kinase isozyme first among 34,873 genes. Applied to a GLGC LDL-cholesterol GWAS, it exposed a regime in which the self-contained null is false almost everywhere: 11.2% of genes reach genome-wide significance, essentially every pathway is significant, and exact FWER control does not rescue the ranking. We argue this is a property of self-contained testing at very large n, not a failure of multiple-testing correction.

Availability: CATFISH is an R package available at `https://github.com/nirwantandukar/CATFISH` under the MIT licence. All simulation, analysis and figure code for this paper is in the `catfish2/` directory of that repository.

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Supplementary information: Supplementary data are available at Bioinformatics online.

Keywords genome-wide association study, pathway analysis, gene-set test, family-wise error rate, Westfall–Young, permutation, ACAT, MAGMA

Abbreviations GWAS, MAGMA, LD, FWER, WY, MVN, ACAT

Key Messages

- Pathway p-values are correlated because pathways share genes; Bonferroni and Benjamini–Hochberg corrections applied to them are therefore miscalibrated, and the direction of the error depends on the overlap structure of the particular pathway database used.
- A Westfall–Young max-statistic null, built by propagating one genome-wide correlated draw through the whole pathway family, controls the family-wise error rate exactly under arbitrary between-pathway dependence and requires no independence assumption anywhere.
- Implementing the omnibus in log space is a correctness requirement, not an optimisation: at GWAS sample sizes above roughly one million the analytic omnibus underflows double precision and the pathway ranking collapses into ties.
- Self-contained pathway tests saturate at very large n. When 11% of all genes are genome-wide significant, a random 50-gene set contains several of them, almost every pathway rejects, and exact FWER control does not restore discrimination. Saturation is a property of the hypothesis being tested, not of the correction applied to it.

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1 Introduction

Genome-wide association studies (GWAS) have transformed complex trait research by identifying thousands of loci linked to disease, development, and quantitative phenotypes. Yet most variants explain only a small fraction of phenotypic variance and often fall in noncoding regions, limiting single-marker mechanistic interpretation (Manolio et al., 2009; Boyle et al., 2017). Complex traits are often highly polygenic, with relevant signals distributed across many modest-effect loci rather than concentrated in a few genome-wide significant variants (Akula et al., 2025; Lappalainen et al., 2024). Gene-set analysis, including pathway-based approaches, is therefore a key post-GWAS strategy for translating summary statistics into biological hypotheses. By aggregating evidence across predefined groups of functionally related genes, pathway-based methods enhance interpretability, reduce the multiple-testing burden, and reveal coordinated pathway disruptions that single-locus analyses may miss (Mooney and Wilmot, 2015; de Leeuw et al., 2015). This is especially valuable for polygenic traits, where many genes may influence the same phenotype through shared biological processes. In such settings, a candidate gene becomes more compelling when locus- or gene-level association evidence is reinforced by pathway-level enrichment from a biologically relevant process. Pathway support therefore provides functional context for prioritization, although it is best interpreted as complementary rather than definitive evidence because pathway annotations remain incomplete and often overlapping.

A plethora of pathway analysis tools has been developed for GWAS. Classical over-representation methods test whether predefined pathways are enriched among significant genes, as implemented in DAVID, g:Profiler, and WebGestalt (Sherman et al., 2022; Kolberg et al., 2023; Liao et al., 2019). Ranking-based methods such as Gene Set Enrichment Analysis (GSEA) assess whether pathway genes are non-randomly concentrated at the top or bottom of a ranked gene list (Subramanian et al., 2005). GWAS-focused extensions, including MAGENTA, INRICH, and MAGMA, add features such as SNP-to-gene mapping, pathway-level modeling, and adjustment for linkage disequilibrium (LD) or gene size (Segrè et al., 2010; Lee et al., 2012; de Leeuw et al., 2015). Accounting for LD is essential in GWAS pathway analysis because nearby variants are often correlated and therefore do not provide independent evidence of association. If LD is ignored, clusters of linked SNPs can be effectively counted multiple times, leading to inflated gene-level aggregation and enrichment biases toward larger genes or regions with dense marker coverage. Several widely used post-GWAS methods therefore explicitly account for LD through model-based or reference-based dependence estimates rather than assuming independence. VEGAS uses multivariate normal simulation, Pascal uses LD from a reference population to compute gene and pathway scores, and MAGMA models gene-level dependence through an approximated correlation structure (Liu et al., 2010; de Leeuw et al., 2015; Lamparter et al., 2016; Watanabe et al., 2017).

These methods address dependence *within* a pathway. A second source of dependence operates *between* pathways and is far less often handled. Curated pathway databases are built from overlapping annotations: a gene that participates in several metabolic steps is a member of several pathways, and hierarchical resources such as Reactome place parent and child pathways in the same family. The pathway-level p-values returned by any method are therefore correlated with one another, sometimes strongly. Nonetheless, the final multiple-testing step in essentially every published pipeline is a Bonferroni or Benjamini–Hochberg adjustment of those p-values, both of which are derived under independence or under positive-regression dependence conditions that overlapping gene sets do not generally satisfy. The resulting family-wise error rate is not the nominal one, and the size of the discrepancy is a property of the particular database rather than of the method, which makes it invisible to the kind of simulation study that generates independent pathways.

The classical solution to exactly this problem is the Westfall–Young max-statistic procedure (Westfall and Young, 1993). Rather than correcting each p-value in isolation, one generates a null realization of the *entire* family of tests, records the single most extreme statistic observed in that realization, and repeats. The distribution of that maximum is the reference distribution for the observed minimum, and the resulting threshold controls the family-wise error rate exactly, whatever the dependence structure among the tests, because the dependence is inherited by construction from the way the null realizations are generated. In a GWAS pathway setting this means drawing correlated gene-level null statistics genome-wide, propagating that single draw through every pathway in the family, and recording the minimum pathway p-value. Nothing about the between-pathway correlation needs to be estimated, modelled, or even named.

Here we extend CATFISH (Combining **C**auchy combination test (ACAT), **A**daptive **T**Fisher (soft) test, **F**isher's test, m**I**n-P, and **S**touffer's method for **H**olistic pathway analysis), a correlation-aware pathway analysis framework that integrates multiple complementary gene-set statistics in a unified omnibus strategy, with Westfall–Young family-wise calibration. Different pathway tests are sensitive to different genetic architectures: some are best at detecting broad, distributed enrichment, whereas others are more sensitive to sparse strong signals or heterogeneous evidence. Instead of relying on a single summary statistic, CATFISH evaluates each pathway with several complementary tests and combines them in an omnibus framework. This improves robustness when the true signal configuration is unknown while preserving the interpretability of the component tests. Beyond pathway-level inference, CATFISH prioritizes candidate genes by integrating locus-level GWAS signals, gene-level MAGMA results, and pathway-level support.

This paper makes three contributions beyond the original CATFISH framework. First, we add the Westfall–Young layer described above and characterize how it behaves: the same correlated null, read down its columns, supplies each pathway's marginal p-value, and read across its rows, the family-wise one. Second, we show that the entire statistic must be computed in log space. This sounds like an implementation detail and is not: at the sample sizes now routine in human GWAS consortia, the analytic omnibus p-value for a typical pathway falls below the smallest representable double-precision number, every such pathway is assigned the same floor value, and the ranking that the whole analysis exists to produce is destroyed without any error being raised. Third, we report a negative result we believe is general. In a GLGC LDL-cholesterol GWAS of roughly 1.3 million individuals, the self-contained null hypothesis is false for almost every gene set that could be constructed, nearly every pathway is significant, and exact family-wise error control does not recover a usable ordering. We interpret this as saturation of the self-contained hypothesis rather than as a defect of the correction,

Table 1 Summary of pathway signal archetypes and expected CATFISH component sensitivity. Each archetype represents a conceptual ranked gene-level association profile rather than an empirically learned class of real pathways.

Archetype	Gene-level p-value pattern	Ranked-profile shape	Expected sensitive tests
Strong Sparse Signal (SSS)	A small number of genes have extremely small p-values, while most genes remain close to null.	Sharp early peak followed by a rapid drop; pronounced elbow.	ACAT, minP
Moderate Dense Signal (MDS)	Many genes show modest but consistent association, without one clearly dominant gene.	Broad shoulder of moderate signals across a substantial fraction of the pathway.	Fisher, Stouffer, soft TFisher
Weak Diffuse Enrichment (WDE)	Most genes are individually non-significant, but the pathway shows a subtle overall shift toward smaller p-values.	Gentle pathway-wide deflection with few or no strong individual peaks.	Stouffer, Fisher
Mixed Signal Architecture (MSA)	One or a few strong genes occur together with a broader set of moderate supporting signals.	Strong leading peak plus a moderate shoulder.	soft TFisher, Fisher, ACAT
Single-Gene Proxy (SGP)	One dominant gene accounts for most of the pathway-level signal, while the remaining genes are near null.	Single extreme peak followed by an otherwise null-like profile.	minP, ACAT

and discuss what it implies for the choice between self-contained and competitive testing as GWAS sample sizes continue to grow.

CATFISH is implemented as an R package and is freely available at `https://github.com/nirwantandukar/CATFISH` under the MIT licence.

1.1 Pathway signal archetypes

CATFISH was developed to capture heterogeneous pathway-level signal patterns, ranging from highly sparse gene-level associations to weak, broadly distributed enrichment. We define five conceptual pathway signal archetypes to clarify how different gene-level profiles are expected to favor different component tests. These archetypes are not intended as empirically learned classes of real pathways but as statistical templates for interpreting test behavior and motivating the complementary design of CATFISH.

For interpretive clarity, we consider a pathway-level working null and alternative. Under the null, gene-level evidence within a pathway matches the background distribution after accounting for dependence, with no systematic excess of small p-values. Under the alternative, the pathway shows enrichment through one or more forms of excess association evidence, including extreme individual genes, broad moderate signals, weak diffuse shifts, or mixed patterns. Figure 1 illustrates the five ranked signal shapes, and Table 1 summarizes their statistical profiles and expected CATFISH component sensitivity.

In short, Strong Sparse Signal (SSS) refers to pathways driven by a few strongly associated genes that favor extreme-tail methods. Single-Gene Proxy (SGP), a special case of SSS, is composed of a signal dominated by one leading gene, but it should be interpreted cautiously as gene-centric. Moderate Dense Signal (MDS) has many modest, consistent gene associations that favor aggregation methods. Weak Diffuse Enrichment (WDE) constitutes subtle pathway-wide shifts, with few individually significant genes, favoring distributed-signal methods. Mixed Signal Architecture (MSA) represents one or a few strong signals plus broader moderate support that favor balanced methods.

2 Methods

2.1 Gene-level inputs and component tests

For a pathway S containing $m = |S|$ genes, let p_g denote the gene-level association p-value for gene $g \in S$ and Z_g the corresponding gene-level Z-statistic. We write $\mathbf{p}_S = (p_1, \ldots, p_m)$ and $\mathbf{Z}_S = (Z_1, \ldots, Z_m)$. Gene-level statistics are obtained with MAGMA (de Leeuw et al., 2015), which aggregates SNP-level GWAS evidence across variants assigned to a gene while accounting for LD, gene size, and local marker density, and which additionally returns the gene–gene correlation matrix induced by LD. These quantities are the common inputs for every CATFISH component test and for both calibration layers.

Each pathway is evaluated with a panel of five complementary component tests (Table 2): the Cauchy combination test (ACAT) (Liu et al., 2019; Liu and Xie, 2020), Fisher's method (Fisher, 1932), an adaptive soft-thresholding TFisher statistic (Zhang et al., 2020), the minimum p-value (minP), and Stouffer's inverse-normal combination (Stouffer et al., 1949). These five were chosen because together they span the archetypes of Section 1.1 and are applicable directly to gene-level p-values or Z-scores without external weights or pathway-specific tuning. The component p-values are aggregated into a single omnibus statistic using an ACAT-type cross-method operator $\mathcal{O}(\cdot)$. The omnibus is not an additional gene-level test but a pathway-level summary of the evidence produced by the component tests for the same pathway.

The soft TFisher component is adaptive over a grid of truncation thresholds τ. In this work we use $\tau \in \{10^{-1}, 5 \times 10^{-2}, 10^{-2}, 5 \times 10^{-3}, 10^{-3}\}$ and take the minimum calibrated TFisher p-value across the grid. This grid is deliberately coarser and shifted upward relative to the original CATFISH defaults. A threshold far into the tail, such as $\tau = 10^{-5}$, is degenerate for pathways of realistic size: the expected number of genes surviving truncation is $m\tau$, so for $m = 50$ fewer than one null draw in a thousand contributes any non-zero term at all and the statistic is identically zero for the rest, which makes the empirical null a point mass and the calibrated p-value uninformative. The grid above keeps the expected number of surviving genes above one across the pathway sizes we consider.

2.2 Two sources of dependence

CATFISH statistics are correlated at two levels, and the two require different treatments.

Within a pathway, the component tests are not independent of one another for two reasons. Each is a deterministic function of the same vector $\mathbf{p}_S$, which induces cross-method dependence even if genes were independent; and the gene-level evidence is itself correlated through LD, overlapping SNP-to-gene assignments, and shared local genomic architecture. Both are handled by the correlated-null calibration of

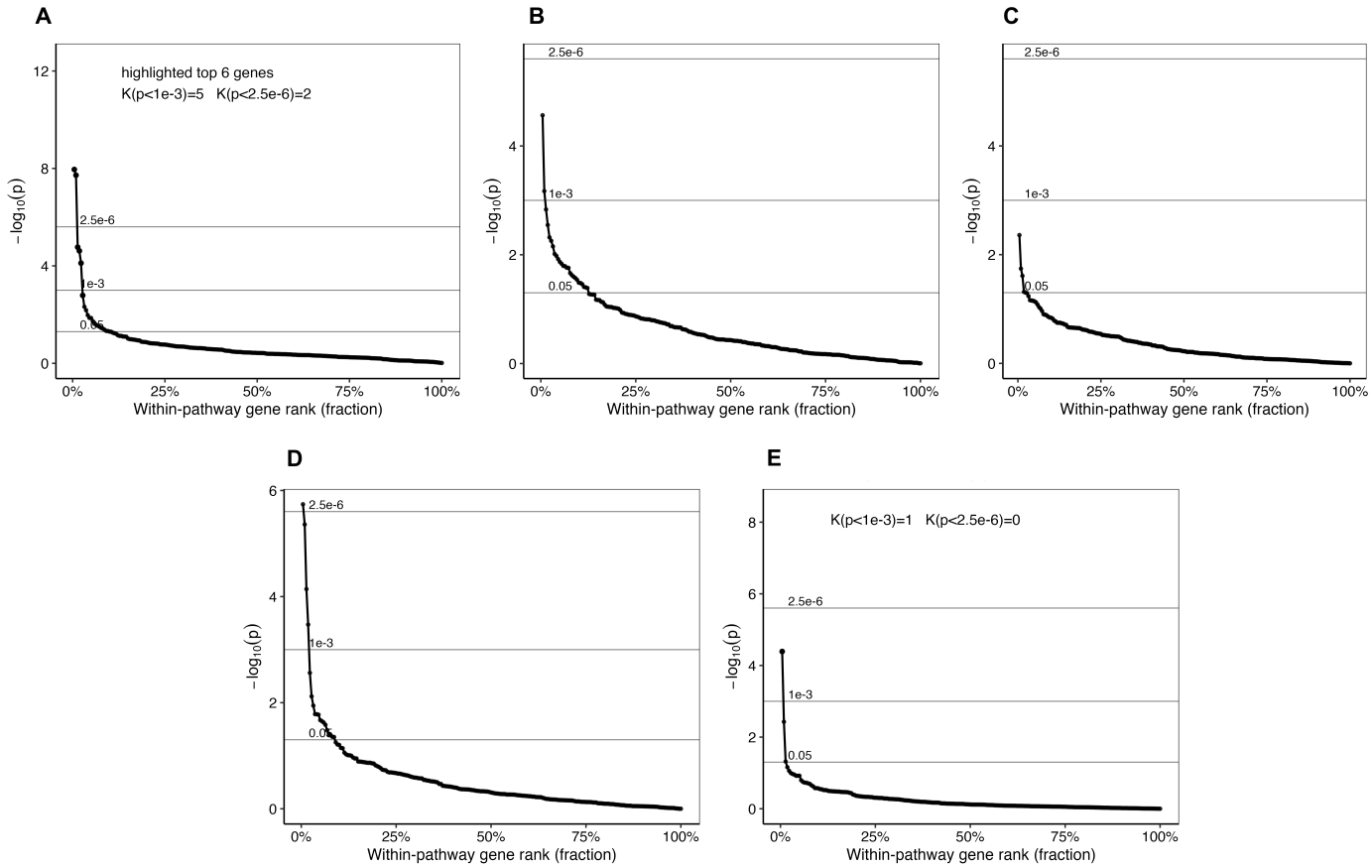


Figure 1 Ranked gene-level association profiles for five pathway signal archetypes. Panels illustrate conceptual within-pathway $-\log_{10}(p)$ profiles for Strong Sparse Signal (SSS), Moderate Dense Signal (MDS), Weak Diffuse Enrichment (WDE), Mixed Signal Architecture (MSA), and Single-Gene Proxy (SGP). Horizontal lines mark representative significance thresholds. These archetypes motivate the use of complementary CATFISH component tests because different methods are preferentially sensitive to different ranked signal shapes.

Table 2 CATFISH pathway-level component statistics. For a pathway S with $m = |S|$ genes and gene-level p-values p_g, each component emphasizes a different pathway signal pattern. The analytic reference distributions define the statistics; final inference uses the dependence-aware calibrations of Section 2.3.

Component	Pathway statistic	Analytic reference	Primary sensitivity
ACAT	$T_{\mathrm{ACAT}}(S) = \frac{1}{m}\sum_{g\in S}\tan\{\pi\,(\frac{1}{2} - p_g)\}$	$p_{\mathrm{ACAT}}(S) = \frac{1}{2} - \frac{1}{\pi}\arctan T_{\mathrm{ACAT}}(S)$	Sparse or extreme-tail signals.
Fisher	$T_{\mathrm{F}}(S) = -2\sum_{g\in S}\log p_g$	Under independence, $T_{\mathrm{F}} \sim \chi^2_{2m}$.	Broad enrichment with many small-to-moderate p-values.
Soft TFisher	$W(S;\tau) = \sum_{g\in S}[-2\log p_g + 2\log\tau]_+$	Adaptive over a τ grid; minimum calibrated p-value is taken.	Intermediate signals concentrated in a subset of genes.
Stouffer	$Z_{\mathrm{St}}(S) = \frac{1}{\sqrt{m}}\sum_{g\in S} Z_g, \quad Z_g = \Phi^{-1}(1-p_g)$	$p_{\mathrm{St}}(S) = 1 - \Phi(Z_{\mathrm{St}}(S))$.	Weak, diffuse, pathway-wide enrichment.
minP	$T_{\min}(S) = \min_{g\in S} p_g$	Under independence, $p_{\min} = 1-(1-T_{\min})^m$.	Single strongest gene or highly sparse signals.

Section 2.3, which recomputes every component from a common dependence-preserving null draw.

Between pathways, the pathway-level p-values are correlated because pathways share genes. In CornCyc, a maize metabolic database, a gene assigned to a branch point of amino-acid metabolism is a member of several biosynthetic pathways at once; in Reactome, parent and child pathways are nested by construction. A gene whose null draw happens to be extreme therefore pushes every pathway containing it in the same direction simultaneously. A calibration that evaluates each pathway against its own null in isolation cannot see this. Scoring the whole family on one shared draw, as Section 2.3 describes, does.

The validity of the correlated-null calibration rests on three assumptions. (i) *Approximate normality*: under the null the pathway-level vector of gene statistics is approximately $\mathbf{Z}_S \sim \mathcal{N}(0, R_S)$. (ii) *Correlation-matrix adequacy*: the correlation matrices built from MAGMA's reported gene pairs are an LD-informed working approximation to the true gene-level covariance, which is not directly observed, and unreported pairs are set to zero; practical adequacy is assessed indirectly through the calibration and robustness analyses below. (iii) *No residual structural confounding*: gene-level inputs have been adjusted so that pathway signals are not driven primarily by gene size or SNP density.

2.3 Calibration against a correlated null: one draw, two readings

CATFISH calibrates every pathway against an empirical null built from a single correlated draw of gene-level statistics, and both the per-pathway and the family-wise statements are read off that same null. There is no separate correction step applied afterwards.

Building the null.

Gene–gene correlations are read from MAGMA's `.genes.raw` output as a list of gene pairs and their correlations. For each chromosome c containing m_c genes we form an $m_c \times m_c$ correlation matrix R_c by starting from the identity and filling in the reported pairs,

$$R_c = I_{m_c} + \sum_{(i,j)\in\mathcal{P}_c} r_{ij}\left(e_i e_j^\top + e_j e_i^\top\right),$$

where $\mathcal{P}_c$ is the set of pairs MAGMA reports for that chromosome. Pairs absent from $\mathcal{P}_c$ are left at zero. MAGMA computes correlations only for gene pairs close enough to share LD, so R_c is banded by construction and a zero between two distant genes on the same chromosome is a good approximation rather than an imputation; a zero arising because a nearby pair was not reported is a genuine approximation error, and we return to it in Section 5. Genes on different chromosomes are treated as independent, which is structurally correct rather than imputed. If R_c is not positive definite we repair it by flooring its eigenvalues at 10^{-6}, reconstructing, and renormalising the diagonal; the implementation reports which chromosomes required this.

Writing $R_c = U_c^\top U_c$ for the Cholesky factor, replicate b draws

$$\mathbf{Z}_c^{(b)} = U_c^\top \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, I_{m_c}),$$

independently per chromosome, and the concatenation $\mathbf{Z}^{(b)}$ is one genome-wide null realisation carrying the estimated LD structure. Gene-level null p-values follow by the Gaussian copula $p_g^{(b)} = 1 - \Phi(Z_g^{(b)})$, matching MAGMA's one-sided sign convention.

Scoring every pathway on every draw.

For each replicate we recompute all five component statistics and the omnibus operator $\mathcal{O}(\cdot)$ for *every* pathway in the family $\mathcal{F} = \{S_1, \ldots, S_K\}$, giving a $B \times K$ matrix of null omnibus values

$$Q_k^{(b)} = \mathcal{O}\left(\mathbf{p}^{(b)}(S_k)\right), \qquad b = 1, \ldots, B, \quad k = 1, \ldots, K.$$

Because every component is recomputed from the same draw, the matrix preserves within-pathway gene dependence, cross-method dependence, and—because all K pathways share the draw—between-pathway dependence as well. The observed values $q_k^{\mathrm{obs}} = \mathcal{O}(\mathbf{p}^{\mathrm{obs}}(S_k))$ are computed identically from the real data. Applying $\mathcal{O}$ to the observed components with no resampling defines the uncalibrated analytic omnibus $\mathrm{Omnibus}_{\mathrm{An}}(S_k) = q_k^{\mathrm{obs}}$, which assumes gene independence and serves as the baseline throughout.

Reading 1 (down a column): the marginal p-value.

Column k of Q is the null distribution of pathway S_k under its own LD. The LD-aware marginal p-value is the Monte Carlo tail probability

$$\tilde{p}_k = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left(Q_k^{(b)} \le q_k^{\mathrm{obs}}\right)}{B+1}.$$

This is the quantity a per-pathway multivariate normal calibration produces, obtained here from the genome-wide draw rather than from a pathway-specific submatrix R_{S_k}; the two use identical correlation information, since R_{S_k} is the submatrix of the R_c above restricted to that pathway's genes. We refer to $\tilde{p}_k$ as the MVN-calibrated p-value throughout, and it is what pathways are ranked on.

Reading 2 (across a row): the family-wise p-value.

Row b of Q is a null realisation of the entire family. Recording its minimum,

$$M^{(b)} = \min_{1\le k\le K} Q_k^{(b)},$$

and comparing each observed value against the resulting distribution gives the Westfall–Young adjusted p-value (Westfall and Young, 1993)

$$\tilde{p}_k^{\mathrm{WY}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left(M^{(b)} \le q_k^{\mathrm{obs}}\right)}{B+1},$$

and rejecting every pathway with $\tilde{p}_k^{\mathrm{WY}} \le \alpha$ controls the family-wise error rate at level α. The between-pathway correlation is never estimated or modelled: because all K pathways are evaluated on the same draw, a gene shared by several pathways perturbs all of them together, exactly as it does in the observed data, and that dependence is inherited by the distribution of M. The quantity $\hat{t}_\alpha = \mathrm{quantile}_\alpha(M)$ is the family-wise significance threshold implied by the pathway database actually in use; its ratio to the Bonferroni threshold α/K is a direct, interpretable measure of how far from independent that database is, and it costs nothing beyond the draws already taken.

The relationship between the two readings is the ordinary one between a raw and an adjusted p-value. The marginal $\tilde{p}_k$ answers "is this pathway enriched?"; the adjusted $\tilde{p}_k^{\mathrm{WY}}$ answers "would evidence this strong appear anywhere among K pathways by chance?". Both are per-pathway quantities and both are reported.

Size standardisation.

A minimum taken across tests is only meaningful if those tests share a scale. The raw omnibus does not: Fisher's statistic grows with m, so the family minimum would be dominated by the largest sets for a purely mechanical reason. We therefore also report a variant in which each observed value is expressed as a z-score against its own column, $z_k = (q_k^{\mathrm{obs}} - \mu_k)/\sigma_k$ with μ_k, σ_k the mean and standard deviation of column k, before the row minimum is taken. This is a scale correction required by the minimum operation; it is not a correction for pathway-size bias, and in Section 4.1 we show that it does not act as one.

2.4 Numerical implementation in log space

The omnibus statistic must be computed in log space throughout. This is a correctness requirement rather than an optimisation, and the failure mode it prevents is silent.

The smallest positive normalized double-precision number is approximately 2.2×10^{-308}. In a GWAS of $n \approx 1.3$ million individuals, gene-level p-values below 10^{-300} are not exotic: in the GLGC LDL-cholesterol data analysed in Section 3.9, the published p-value column is exactly zero for 473 variants on chromosome 1 alone, including the *SORT1* lead variant rs12740374 at $-\log_{10} p = 1725$. Aggregating even a handful of such genes with Fisher's method produces a pathway statistic whose p-value underflows to zero in double precision. Every affected pathway is then assigned the same floor value, ties are broken arbitrarily by storage order, and the ordering that the entire analysis exists to produce is destroyed without any warning being emitted. In our first implementation this manifested as a top-ranked pathway list that was, on inspection, alphabetical.

Three changes are required. (i) All component statistics are computed on $\ell_g = \log p_g$ and return log p-values, using `stats::pchisq(..., log.p=TRUE)` and `stats::qnorm(..., log.p=TRUE)` rather than their probability-scale equivalents. (ii) The ACAT sum is evaluated with the small-p asymptote. For $p < 10^{-15}$, $\tan\{\pi(\frac{1}{2} - p)\} \to 1/(p\pi)$, so the contribution of those genes is accumulated as a log-sum-exp of $-\ell_g - \log \pi$ and only the remaining genes use the trigonometric form directly; symmetrically, for a large combined statistic T the inversion $p = \frac{1}{2} - \arctan(T)/\pi$ is replaced by $p \to 1/(T\pi)$. (iii) Minimum-p and omnibus inversions use `log1p(-exp(·))` in place of $1 - x$ wherever the argument is close to one. With these changes the pipeline represents pathway p-values down to $e^{-745} \approx 10^{-324}$ and, more importantly, preserves their *ordering* far below the point at which the probability-scale computation collapses. The reference implementation is `catfish2/scripts/00_lib.R`.

2.5 Candidate-gene prioritization

Pathway inference is a means to an end: the output a biologist acts on is a short list of genes. CATFISH produces one by combining three evidence layers into a single composite score. The score is a heuristic ranking device, not a formal inferential model, and we are explicit about that: it has no null distribution and no p-value, and its purpose is to order genes for follow-up.

2.5.1 Rank normalization

Each layer is placed on a common scale by rank rather than by its raw statistic, so that layers with very different distributions can be added. For gene g let $r_{\mathrm{GWAS}}(g)$ be its rank by minimum SNP-level p-value among genes with at least one assigned variant, $r_{\mathrm{MAGMA}}(g)$ its rank by MAGMA gene-level p-value among all genes with a result, and $r_{\mathrm{PATH}}(g)$ the rank of g's best pathway among the N_{path} tested. Rank one denotes the strongest evidence. The rank scores are

$$S_{\mathrm{GWAS}}(g) = -\log_{10}(r_{\mathrm{GWAS}}(g)/N_{\mathrm{gene}}),$$
$$S_{\mathrm{MAGMA}}(g) = -\log_{10}(r_{\mathrm{MAGMA}}(g)/N_{\mathrm{gene}}),$$
$$S_{\mathrm{PATH}}(g) = -\log_{10}(r_{\mathrm{PATH}}(g)/N_{\mathrm{path}}),$$

each lying in $[0, \log_{10} N]$. Genes with no assigned variant receive $S_{\mathrm{GWAS}} = 0$ and genes in no annotated pathway receive $S_{\mathrm{PATH}} = 0$, so an absent layer neither helps nor penalises beyond contributing nothing.

A rank scale is used rather than $-\log_{10} p$ directly for two reasons. It is invariant to the very different tail behaviour of the three layers, and it is immune to the floors that arise when p-values underflow, provided the underlying quantity still orders correctly—a condition that requires care in practice, as Section 3.9 shows.

2.5.2 Multi-pathway hub bonus

Genes annotated to several pathways receive

$$B_{\mathrm{hub}}(g) = \log_{10}(n_{\mathrm{path}}(g) + 1),$$

where $n_{\mathrm{path}}(g)$ counts distinct pathways containing g. The logarithm keeps connectivity from dominating: a gene in five pathways gains roughly 0.23 score units at the weight below, enough to move it within a tier but not to lift a gene lacking primary genetic support.

2.5.3 Composite score and weights

$$\mathrm{Score}(g) = S_{\mathrm{GWAS}}(g) + S_{\mathrm{MAGMA}}(g) + 0.5\, S_{\mathrm{PATH}}(g) + 0.3\, B_{\mathrm{hub}}(g).$$

Because all terms share the $-\log_{10}(\mathrm{rank}/N)$ scale, the coefficients act as relative importance multipliers. Pathway evidence is halved because a pathway p-value is a *set-level* statistic shared by every member gene: at full weight it would impose the same bonus on all genes in a set and exaggerate gene-level resolution. GWAS and MAGMA receive equal weight as complementary measures of direct genetic evidence; MAGMA's advantage in aggregating multi-variant signal is already expressed in its typically smaller p-values and hence its rank. The hub coefficient 0.3 is deliberately secondary.

One caveat applies whenever both genetic layers derive from the same variants through the same window, which is the usual case. They are then not independent sources of evidence, and a gene supported by "all three layers" has really been supported by two. We report the per-layer ranks alongside the composite score throughout so that this is visible rather than hidden.

2.5.4 Rank-based support under non-significance

For polygenic traits few genes or pathways clear a family-wise threshold, and a score defined only over significant layers would be empty. We therefore define layer support by rank: GWAS support is the top 1% of genes by minimum locus p-value, MAGMA support the top 5% by gene-level p-value, and pathway support membership in any annotated pathway. When significant pathways exist we use only those; otherwise we retain the top twenty pathways or the top 10% by omnibus p-value. These fallbacks keep the layer annotation informative in the regime where the method is most needed.

2.6 Simulation design

All simulations operate at the level of the omnibus statistic and its null. Pathways are defined as overlapping subsets of a single shared gene universe: $G = 3000$ genes arranged in independent blocks of 50 with exchangeable correlation ρ within a block, standing in for chromosomes or LD blocks, from which $K = 150$ pathways of m genes each are drawn at random from the first 600 genes. Each gene therefore sits in several pathways, and one genome-wide draw perturbs every pathway containing it at once, which is the dependence structure a real pathway database has and the one the family-wise procedure exists to handle. Every study below reports both readings of the resulting null matrix: the per-pathway marginal type I error from its columns, and the family-wise error rate from its rows, with Bonferroni as the competitor for the latter. Thresholds are fitted on one set of draws and evaluated on a held-out set, so no rate is correct by construction. We deliberately do not simulate genotypes and phenotypes and push them through a full GWAS; that experiment is valuable and is the main item of future work (Section 5), but it is not what the questions below require, and including a half-done version of it would obscure which conclusions rest on which assumptions. Each study below corresponds to one script in `catfish2/scripts/`, and each writes a CSV to `catfish2/results/` that the tables and figures are generated from.

1. **Null calibration and type I error** (`01_null_calibration.R`). Under the global null we compare four ways of turning omnibus statistics into decisions: the analytic independence-based p-value; MVN calibration against a shared per-size null bank; a Bonferroni family-wise correction; and the Westfall–Young family-wise threshold obtained from the distribution of the family minimum. We cross three LD regimes, $\rho \in \{0, 0.3, 0.6\}$, with three pathway sizes, $m \in \{5, 20, 50\}$, and report empirical type I error at $\alpha \in \{0.05, 0.01\}$ together with the genomic-control inflation factor $\lambda = \text{median}(\chi^2_1\text{-transformed } p)/0.4549$.
2. **Component-breakage stress test** (`02_breakage_stress.R`). We deliberately miscalibrate k of the five components by replacing each affected component's p-value with p^γ, and ask whether the omnibus and its calibration survive. The direction of the corruption matters and both are run. For $0 < p < 1$, $\gamma < 1$ gives $p^\gamma > p$ and makes the component *conservative*; $\gamma > 1$ gives $p^\gamma < p$ and makes it *anti-conservative*, which is the direction that produces false positives and therefore the one a robustness claim has to address. We use $\gamma = 0.7$ and $\gamma = 1.4$, sweep $k = 0, \ldots, 4$, evaluate up to three distinct component combinations at each k, and cross this with the same LD regimes at $m \in \{20, 50\}$. The corruption is applied to the null replicates as well as to the observed statistics, which is the mechanism by which resampling calibration can absorb it.
3. **Leave-one-out component sensitivity** (`03_adaptive_loo.R`). We remove each component in turn, recompute the null bank and the calibration from scratch, and measure the change in type I error and λ. This isolates how much each component contributes to, or destabilises, the calibrated omnibus, and identifies components whose removal would change inference.
4. **Per-component calibration** (`09_component_calibration.R`). Rather than combining, we calibrate each of the five components against its own null and report analytic and calibrated type I error, family-wise rate and λ per component. This identifies which statistics LD actually breaks and whether a single calibrated component is usable in place of the omnibus when the signal architecture is known in advance.
5. **Robustness to an incomplete correlation file** (`08_missing_correlatio`). Data are generated from the true correlation structure while the calibration—null bank, marginal p-values and Westfall–Young threshold—is built from a matrix in which a fraction $\{0, 0.25, 0.5, 0.75, 1\}$ of off-diagonal entries has been zeroed, exactly as an unreported gene pair would be. The evaluation draws are shared across missingness levels so that only the calibration side varies. For real data the companion script `10_pair_coverage.R` measures where a given correlation file sits on this axis, by reporting the fraction of within-pathway gene pairs that are on the same chromosome and close enough to be correlated for which MAGMA reports a value.
6. **Ranking and family-wise thresholds** (`04_wy_vs_mvn.R`). Each replicate contains 300 pathways of which 15 carry a latent mean shift $\mu = 1.5$ on the gene-level z scale. We measure the median within-replicate rank of the truly enriched pathways under the analytic and MVN orderings against the random expectation of 150, and we compute the Westfall–Young family-wise threshold and its ratio to the Bonferroni threshold $0.05/K$. The mean shift is applied at the pathway level and is not a simulated GWAS; it quantifies ordering behaviour and threshold calibration, not power for any particular genetic architecture.

Unless stated otherwise, each configuration uses B null draws for the calibration bank and a separate set of replicate datasets for evaluation, with both values set at the top of each script so that a reduced smoke-test configuration can be run with `bash run_all.sh quick`.

2.7 Real-data analyses

Maize grain nitrogen. Gene-level statistics were computed with MAGMA from a mixed-linear-model (MLM) GWAS of grain nitrogen content, using a 10 kb upstream and downstream window for SNP-to-gene assignment. Pathways were taken from CornCyc (Hawkins et al., 2021; Caspi et al., 2008), giving 172 pathways after size filtering. Gene–gene correlations were recovered per chromosome from MAGMA's `.genes.raw` output; we verified explicitly that all ten chromosomes were represented, after an earlier run in which only chromosome 1 correlations had been parsed and 85% of the genome was silently treated as independent. Candidate-gene scoring used the three-layer score of Section 2.5 over all 34,873 annotated genes.

Human LDL cholesterol. We used the Global Lipids Genetics Consortium LDL-cholesterol summary statistics for the European-ancestry meta-analysis ($n \approx 1.3$ million) (Graham et al., 2021).

Variant filtering and the p-value floor. Variants were restricted to rsIDs present in the 1000 Genomes European reference panel used for LD. The released `pvalue` column underflows to exactly zero at the strongest loci, so all probabilities were reconstructed from `pvalue_neg_log10`. For the MAGMA gene analysis, whose `--pval` input requires a probability, these were floored at 10^{-300}; MAGMA is unaffected because the floor is far beyond any gene-level resolution it reports. The candidate-gene GWAS layer is affected, because it ranks genes by their minimum variant p-value and the floor collapses the strongest genes into a single tie. That layer therefore reads an unfloored $-\log_{10} p$ file written by `11_munge_neglog10.sh`; Section 3.9 reports what the floor costs when it is not removed.

Gene analysis. Variants were assigned to genes with a 10 kb upstream and downstream window using the NCBI37.3 gene locations. MAGMA gene analysis was run per chromosome with `--batch` and combined with `--merge`, which preserves the gene–gene correlations in the merged `.genes.raw` file; running the chromosomes as independent jobs and concatenating the outputs would discard them.

Pathways. Pathway inference used the Reactome subset of MSigDB C2:CP (1,390 sets after size filtering), with HumanCyc additionally converted into the CATFISH pathway format for comparison. We report Reactome because a hierarchical database is where between-pathway dependence is strongest, which is the regime the family-wise procedure is designed for.

Candidate genes. The GWAS layer uses MAGMA's own `.genes.annot` file rather than re-deriving a window, so that the GWAS and gene layers are assigned from identical variant sets. The pathway layer is ranked on the observed omnibus statistic rather than the Westfall–Young adjusted p-value. The two give the identical ordering, since the adjusted p-value is a monotone transform of the statistic, but the adjusted value is floored at $1/(B+1)$ and the entire top of the list is tied there at the B used here.

Positive controls. Thirteen genes with established roles in LDL-C biology—*LDLR*, *PCSK9*, *APOB*, *APOE*, *SORT1*, *HMGCR*, *NPC1L1*, *ABCG8*, *LPA*, *CETP*, *LDLRAP1*, *MYLIP* and *ANGPTL3*—were named before the prioritization was run and are used to assess it. They are a validation set, not a discovery; the score has no access to them.

3 Results

3.1 Independence-based calibration is anti-conservative; MVN and Westfall–Young are not

Under the global null, the analytic omnibus p-value is valid only if the genes within a pathway are independent. Table 3 and Figure 2 show what happens when they are not. At $\rho = 0$ and $m = 5$ the analytic type I error was 0.0362 against a nominal 0.05, which is the behaviour one expects when the generating assumption holds. As either the within-pathway correlation or the pathway size increases, the analytic calibration departs from nominal, reaching 0.3386 at $\rho = 0.6$, $m = 50$; the worst case across the grid was 0.3386. The direction is the expected one: correlated genes carry less independent information than the analytic reference distribution assumes, so the statistic looks more extreme than it is.

MVN calibration against a pathway-specific correlated null removed most of this. Empirical type I error was 0.0472 at $\rho = 0$, $m = 5$ and 0.0425 at $\rho = 0.6$, $m = 50$, remaining within $[0.0400, 0.0568]$ across the whole grid. At the stricter level $\alpha = 0.01$, the analytic calibration reached 0.3011 against the MVN calibration's 0.0157, indicating that the discrepancy grows rather than shrinks in the tail, which is where pathway analysis operates.

The family-wise columns behave as theory predicts. The Westfall–Young threshold delivered empirical family-wise error between 0.0500 and 0.0500 against the nominal 0.05 in every configuration, which is the defining property of the procedure: the threshold is *defined* as a quantile of the realized family minimum, so it is correct by construction whatever the dependence, and the simulation serves as a check on the implementation rather than as evidence for the method. The Bonferroni column, by contrast, is correct only in the independent configurations and becomes conservative as ρ grows, since the effective number of independent tests falls below K.

3.2 Genomic-control inflation tracks the same failure

The genomic-control inflation factor λ summarises the same phenomenon in a form directly comparable to GWAS practice (Figure 3). For the analytic omnibus, λ ranged from 0.115 to 1.065 across the grid, increasing with both ρ and m. After MVN calibration it fell to $[0.925, 1.181]$. Because λ is computed from the median of the null distribution rather than its tail, it is a conservative diagnostic here: it understates the problem relative to the type I error at $\alpha = 0.01$. A pathway analysis reporting λ near one is therefore not by itself evidence that the tail is well calibrated.

3.3 The omnibus degrades gracefully when its components are broken

A multi-test omnibus is only useful if a single misbehaving component cannot dominate it, and a family-wise procedure is only useful if that robustness survives at the family level too. We tested both by corrupting components on purpose, replacing the p-value of each of k selected components with p^γ and running the conservative ($\gamma = 0.7$) and the anti-conservative ($\gamma = 1.4$) direction over the overlapping pathway universe of Section 2.6, in which each gene belongs to 8.8 pathways on average (Table 4, Figure 4).

With no components corrupted ($k = 0$), the analytic omnibus had mean marginal type I error 0.0788 and the calibrated omnibus 0.0475; at the family level the Westfall–Young rate was 0.0547 and Bonferroni 0.1290. Under anti-conservative corruption of four of the five components ($k = 4$, $\gamma = 1.4$) the analytic omnibus degraded to 0.1678, while the calibrated marginal rate held at 0.0478 and the Westfall–Young family-wise rate at 0.0549 against Bonferroni's 0.3111. Under conservative corruption the analytic omnibus moved the other way, to 0.0349, losing power rather than control, while the calibrated marginal and family-wise rates held at 0.0471 and 0.0529. Across both directions and every configuration the family-wise rate stayed within $[0.0420, 0.0673]$ and the marginal rate within $[0.0336, 0.0536]$. Genomic-control inflation for the same runs (Figure 5) tracks the analytic statistic's drift in both directions while staying near one for the calibrated statistic.

The interpretation is specific and worth stating carefully. Resampling calibration is robust to component miscalibration because the *same* corruption is applied to the null replicates as to the observed data, so a component that is uniformly miscalibrated shifts the null and the observation together and largely cancels; the family-wise threshold, being a quantile of the same corrupted null, moves with them. This is a real and useful property, but it protects against systematic miscalibration, not against a component that is wrong in a way that correlates with the signal of interest. A component whose error depended on the data would not cancel, and we have not tested that case. The analytic calibration has no such protection and inherits the corruption directly, in whichever direction it was applied.

3.4 No single component is load-bearing

Removing each component in turn and rebuilding the null bank, the marginal calibration and the family-wise threshold from scratch (Table 5, Figure 6) left the marginal type I error within $[0.0326, 0.0539]$ against 0.0475 for the full five-component omnibus,

Table 3 Empirical type I error and genomic-control inflation for the CATFISH omnibus under the global null, by within-pathway LD level and pathway size. Analytic: independence-based omnibus. MVN: calibrated against a pathway-specific correlated null. Bonferroni and Westfall–Young (WY) columns are family-wise error rates across the pathway family. Nominal level $\alpha = 0.05$.

LD	ρ	m	Analytic	MVN	Bonferroni	WY	λ_{an}	λ_{mvn}
independent	0.0	5	0.0362	0.0472	0.0400	0.0500	0.843	0.934
independent	0.0	20	0.0383	0.0474	0.0360	0.0500	0.965	0.993
independent	0.0	50	0.0392	0.0556	0.0240	0.0500	1.065	0.925
moderate	0.3	5	0.0847	0.0523	1.0140	0.0500	0.730	1.002
moderate	0.3	20	0.2031	0.0568	13.7300	0.0500	0.712	1.060
moderate	0.3	50	0.2857	0.0400	32.5300	0.0500	0.862	0.998
strong	0.6	5	0.1238	0.0542	3.7500	0.0500	0.550	1.181
strong	0.6	20	0.2599	0.0473	26.9300	0.0500	0.115	1.031
strong	0.6	50	0.3386	0.0425	47.8840	0.0500	0.120	1.002

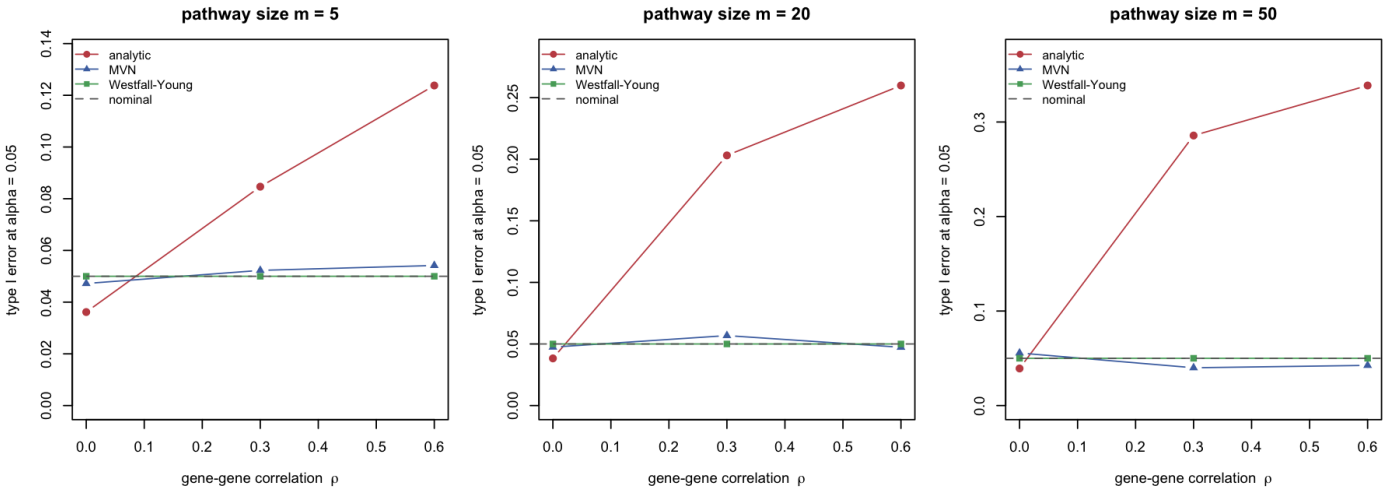


Figure 2 Empirical type I error of the CATFISH omnibus under the global null at $\alpha = 0.05$, by within-pathway LD level ($\rho = 0,\ 0.3,\ 0.6$) and pathway size ($m = 5,\ 20,\ 50$). The horizontal line marks the nominal level. The analytic independence-based calibration departs from nominal as either correlation or pathway size increases; MVN calibration against a pathway-specific correlated null, and the Westfall–Young family-wise threshold, both track the nominal level.

and the Westfall–Young family-wise rate within $[0.0400, 0.0680]$ against 0.0547. The largest departure from nominal followed removal of `minP` (0.0543).

The same table carries the sharpest contrast in the paper between the two family-wise procedures. Averaged over pathway sizes and leave-one-out variants, the Westfall–Young rate was 0.0587, 0.0495 and 0.0537 in the independent, moderate and strong LD regimes—flat at nominal—while the Bonferroni rate moved from 0.0227 to 0.1460 to 0.1997, reaching 0.2640 at worst. We are careful about what this does and does not show. Bonferroni is applied here to the analytic omnibus p-value, which is itself anti-conservative under LD (Section 3.1), so the inflation is the compounding of an uncalibrated per-test statistic with a correction that cannot repair it, which is what a pipeline that skips calibration actually does. Dependence alone would make Bonferroni *conservative*, and that is what the independent rows show. The lesson is that a family-wise correction inherits whatever miscalibration the per-test statistic carries, whereas the Westfall–Young threshold, being a quantile of the same null the statistic came from, does not.

Within that pattern, removing Fisher or Stouffer lowered the Bonferroni rate noticeably while removing ACAT, minP or TFisher did not, which is consistent with the two aggregation-type components being the ones most inflated by within-pathway correlation.

No single deletion broke error control at either level, which is the expected consequence of calibrating empirically: the null bank is recomputed for whichever components remain, so both the marginal p-value and the family-wise threshold remain valid for any subset. What changes with the subset is sensitivity, not validity. The practical implication is that users may drop a component they

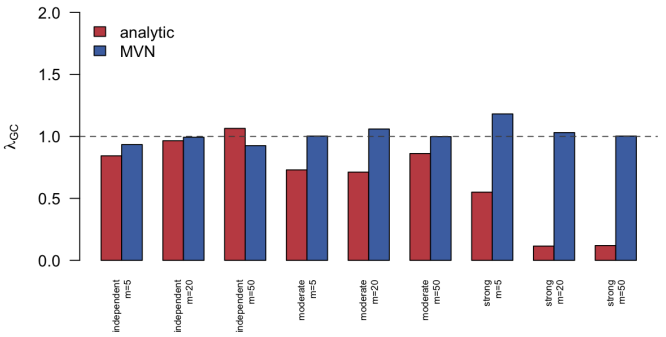


Figure 3 Genomic-control inflation factor λ for the analytic and MVN-calibrated omnibus under the global null, by LD level and pathway size. The horizontal line marks $\lambda = 1$.

distrust for a particular dataset—for example minP when gene-level p-values have been floored at a reporting limit—without invalidating the inference, at some cost in power against the architectures that component is best at.

Genomic-control inflation behaves the same way: the fourth panel of Figure 6 shows λ for the calibrated marginal statistic staying near one for every deletion, so no component is load-bearing for the inflation diagnostic either.

3.5 Linkage disequilibrium breaks the aggregation tests, not the extreme-value tests

The omnibus hides which of its inputs is failing. Calibrating each component separately (Table 6, Figure 7) shows that the damage LD does is concentrated in two of the five, and that the identity of those two is predictable from what the statistics measure.

Under the independent null every component was correctly sized, from 0.0526 for ACAT to 0.0496 for Stouffer, which is the expected behaviour when the generating assumption holds. Under strong LD the two aggregation statistics—Fisher, which sums $-2 \log p$ across the pathway, and Stouffer, which sums Z-scores—inflated to 0.1566 and 0.1805 against the nominal 0.05. The two extreme-value statistics barely moved: ACAT went to 0.0547 and minP to 0.0452. The mechanism is transparent. An aggregation statistic adds one term per gene and implicitly treats each as independent evidence, so correlated genes are counted repeatedly and the statistic grows; a statistic that depends on the most extreme gene is insensitive to how many near-duplicates of it the pathway contains. The omnibus inherited the inflation of its worst inputs, reaching 0.1371.

Calibration against the correlated null removed this for every component, bringing Fisher to 0.0372, Stouffer to 0.0323 and the omnibus to 0.0342; if anything it over-corrected slightly at strong LD. The practical consequence is that a user who knows a pathway's signal architecture may use a single calibrated component in place of the omnibus without losing error control, which is what makes the architecture-specific testing described in Section 2.3 usable.

This result also explains an observation in Section 3.4 that would otherwise look arbitrary: removing Fisher or Stouffer lowered the Bonferroni family-wise rate noticeably while removing ACAT, minP or TFisher did not. Those are the components that were inflated in the first place.

One implementation note belongs here. The adaptive soft TFisher statistic is undefined on draws where no gene falls below the smallest truncation threshold, which happened on up to 4.0% of draws at strong LD. A single such draw in a calibration bank is enough to make the entire calibrated p-value undefined unless non-finite draws are filtered before the tail count, which the implementation now does.

3.6 Both calibrations degrade when the correlation file is incomplete, and the family-wise one degrades faster

Every result above assumes the gene–gene correlation matrix is the one the data were generated from. In practice it is assembled by filling MAGMA's reported gene pairs into an identity matrix, so a pair MAGMA does not report is silently treated as independent. We measured the cost directly by generating data from the true correlation structure while calibrating from a matrix with a known fraction of its off-diagonal entries zeroed (Table 7, Figure 8).

With a complete matrix the marginal type I error was 0.0501 and the Westfall–Young family-wise rate 0.0625, both at nominal. Dropping half the pairs raised them to 0.0877 and 0.1475; dropping all of them, which is the independence assumption in its pure form, gave 0.1168 and 0.2200 under moderate LD and 0.1658 and 0.2825 under strong LD.

Expressed as multiples of each procedure's own nominal level—the only fair comparison between a per-pathway rate and a per-family one—the marginal calibration was 1.8× nominal at 50% missingness and the family-wise procedure 2.9×. Westfall–Young is therefore roughly twice as sensitive to an incomplete correlation file as the marginal calibration it is built from. The reason is structural rather than incidental: the family minimum over K pathways is a tail statistic, so it depends on the extreme of the null distribution, which is precisely where an error in the correlation structure does the most damage. The marginal p-value depends only on the bulk.

This does not overturn the case for family-wise control, because the comparison that matters is against the correction it replaces rather than against the marginal rate it is not competing with. Bonferroni's family-wise rate in the same runs was 0.1850 and 0.2475 and does not depend on the correlation file at all, since it is computed from the analytic p-value. Westfall–Young remained the better family-wise procedure up to 75% missingness, and lost only when essentially no correlation information remained—at which point it is calibrating against a null that does not resemble the data.

It does, however, change how the procedure should be reported. A Westfall–Young p-value is only as trustworthy as the correlation file behind it, and the completeness of that file is measurable rather than assumed. The quantity to check is narrow: a correlation matrix from MAGMA is *supposed* to be sparse, because correlations are computed only for gene pairs close enough to share LD, so a zero between distant genes is correct rather than missing. What matters is the fraction of within-pathway gene pairs that are on the same chromosome and close enough to be correlated for which a value is actually reported. We provide `10_pair_coverage.R` for this, and recommend that it be run and its result stated alongside any family-wise p-value.

Finally, and consistently with Section 3.2, λ failed to detect any of this. Across the whole missingness sweep it stayed between 0.926 and 1.141 while the marginal type I error more than doubled. This is the second independent setting in which λ near one coexists with a badly miscalibrated tail, and it is the clearest argument we can make that

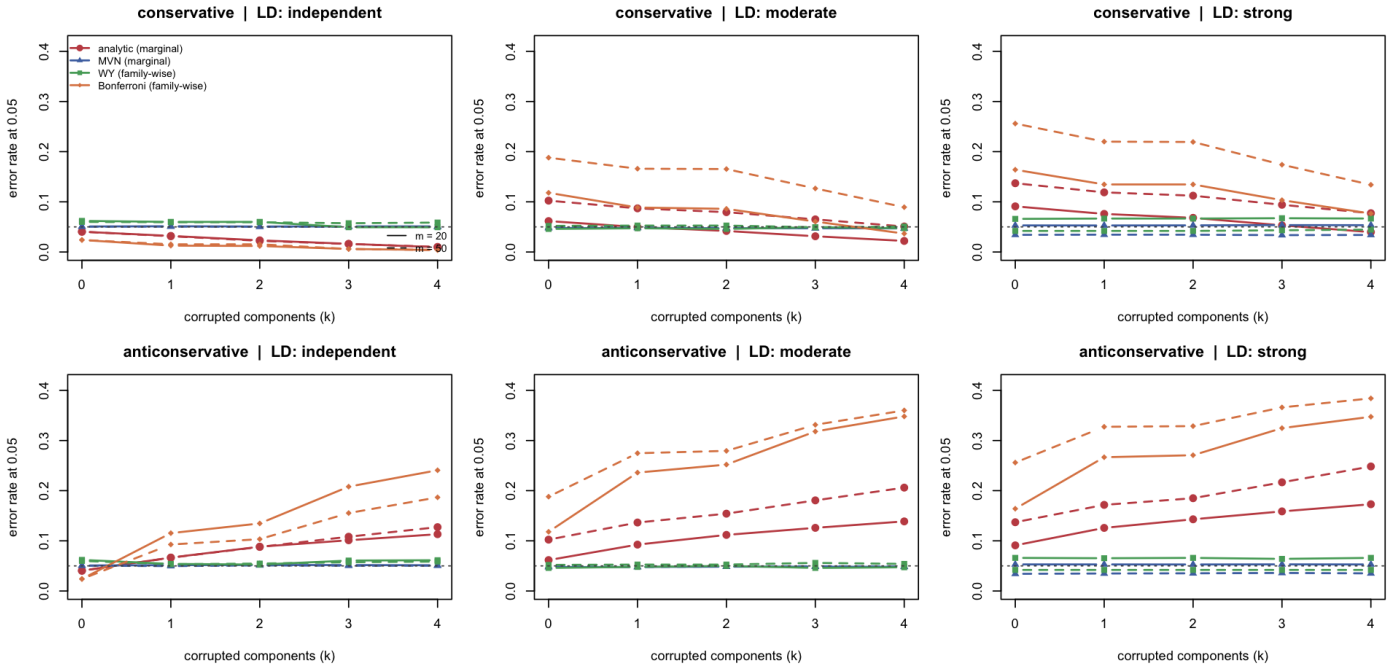


Figure 4 Component-breakage stress test on overlapping pathways. Rows are the direction of the deliberate corruption ($p \mapsto p^{\gamma}$, with $\gamma = 1.4$ anti-conservative and $\gamma = 0.7$ conservative); columns are the within-pathway LD level. Each panel shows the per-pathway marginal rates and the family-wise rates together, since all four are rates at the nominal 0.05 marked by the dotted line. Line type distinguishes pathway size. The analytic statistic and the Bonferroni correction applied to it both drift with k; the calibrated marginal rate and the Westfall–Young family-wise rate do not.

Figure pending: `Fig3b_breakage_lambda.png`
generated by `scripts/05_figures.R`

Figure 5 Genomic-control inflation λ for the component-breakage runs, by corruption direction and LD level. λ is a property of the marginal p-value distribution and has no family-wise counterpart, so it is shown separately from Figure 4. The analytic statistic drifts with k in whichever direction the components were corrupted; the calibrated statistic stays near one.

λ should not be used as the sole calibration diagnostic in pathway analysis.

3.7 Westfall–Young thresholds differ substantially from Bonferroni

Table 8 and Figure 9 quantify the gap between the family-wise threshold implied by the actual dependence structure and the Bonferroni threshold $0.05/K = 1.67 \times 10^{-4}$. Across configurations the Westfall–Young threshold ranged from 1.47×10^{-4} to 3.50×10^{-4}, giving Bonferroni-to-WY ratios between 0.48 and 1.13. A ratio above one means Bonferroni is conservative relative to the exact family-wise threshold, and the degree of conservatism grows with the correlation between tests, as expected.

For the ordering of truly enriched pathways, the median within-replicate rank of the 15 enriched pathways was 11.8 under the analytic ordering and 11.8 under the MVN ordering, against a random expectation of 150. Both orderings recover the enriched pathways; the calibration layer primarily affects which of them clear a threshold, not their relative order. This is worth stating because it bears on how these methods should be used in practice: for a polygenic trait in which nothing is expected to clear a family-wise threshold, the ranking is the deliverable, and the ranking is comparatively robust to the choice of calibration.

3.8 Maize grain nitrogen: Westfall–Young recovers the known pathway and gene

We applied the full pipeline to a mixed-linear-model GWAS of maize grain nitrogen content, testing 172 CornCyc pathways. The leading pathway under the size-standardized Westfall–Young ordering was `HOMOSERSYN-PWY`, L-homoserine biosynthesis (13 genes; omnibus $p = 1.58 \times 10^{-3}$; size-standardized family-wise $p = 0.070$), followed by

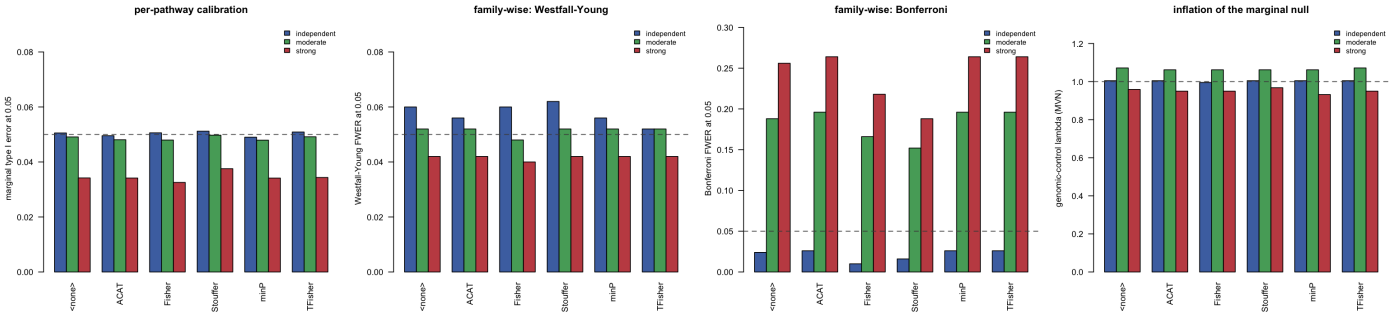


Figure 6 Leave-one-out component sensitivity at $m = 50$, with each component removed in turn and the null bank, marginal calibration and Westfall–Young threshold rebuilt from scratch. From left: marginal type I error, Westfall–Young family-wise error rate, Bonferroni family-wise error rate, and genomic-control inflation λ for the calibrated marginal statistic. The dashed line marks the nominal level (0.05, or $\lambda = 1$). Note the y-scale of the third panel, which is the point of the comparison.

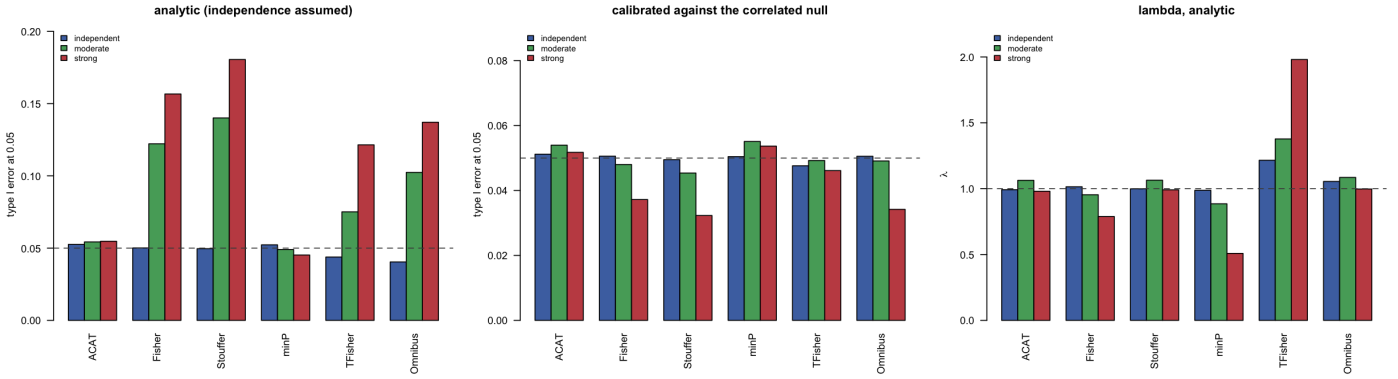


Figure 7 Per-component calibration at $m = 50$. Left: type I error at $\alpha = 0.05$ under the analytic independence assumption. Centre: the same components calibrated against the correlated null. Right: genomic-control inflation λ for the analytic statistics. The aggregation components, Fisher and Stouffer, are the ones LD inflates; the extreme-value components, ACAT and minP, are nearly unaffected. Calibration restores all five.

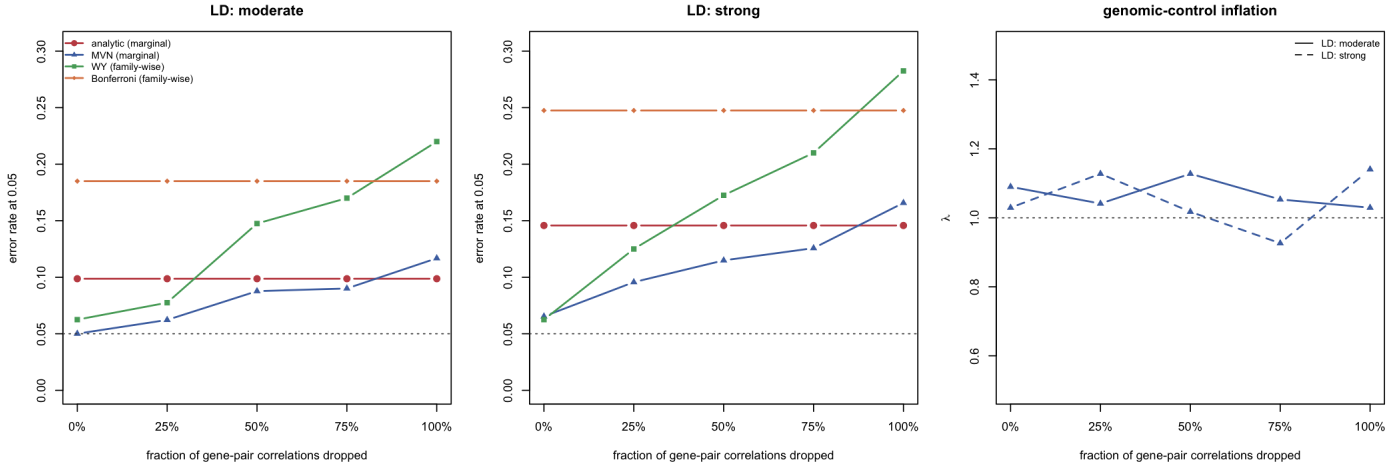


Figure 8 Robustness to an incomplete gene–gene correlation file. Data are generated from the true correlation structure while calibration uses a matrix with the stated fraction of off-diagonal entries zeroed. Left and centre: error rates at $\alpha = 0.05$ under moderate and strong LD. Right: genomic-control inflation λ for the marginal calibration, which stays near one throughout despite the type I error more than doubling.

`PWY-5097`, L-lysine biosynthesis VI (25 genes; omnibus $p = 3.04 \times 10^{-3}$; size-standardized family-wise $p = 0.338$).

This ordering reproduces, independently and from a different statistical framework, the result of a previously reported analysis

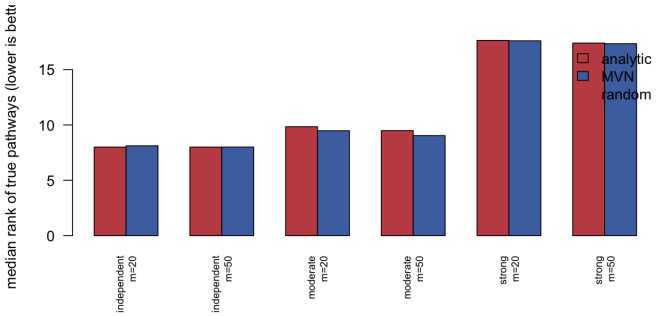


Figure 9 Median within-replicate rank of truly enriched pathways under the analytic and MVN orderings, compared with the random expectation $K/2$, by LD level and pathway size.

of the same trait (Tandukar and Rellán-Álvarez, 2025), in which L-homoserine biosynthesis was the only pathway enriched across all populations examined and L-lysine biosynthesis VI was the runner-up for reasons that analysis sets out. Recovering the same two pathways in the same order out of 172 candidates, using a method that shares no calibration machinery with the original, is the strongest external check we can presently make on the pipeline.

The three-layer candidate score of Section 2.5, computed over all 34,873 annotated maize genes with pathway evidence taken from the Westfall–Young adjusted p-values, placed `Zm00001eb220040` first. This gene is the aspartate kinase isozyme named in the earlier analysis, and L-homoserine biosynthesis was its highest-scoring pathway, so the gene- and pathway-level layers agree rather than merely coexisting.

Two caveats are essential. First, nothing in this analysis is significant: the leading pathway's size-standardized family-wise p-value is 0.070, and under the unstandardized Westfall–Young ordering it is 0.284 against a Bonferroni value of 0.272. The result is a ranking that agrees with prior biology, not a discovery. Second, grain nitrogen is highly polygenic and the MLM GWAS contained no genome-wide significant genes at all (Table 10), which is precisely the regime in which a ranking is the only available output. We regard the agreement as encouraging evidence that the ordering is meaningful in the low-signal regime, and we do not claim more than that.

3.9 Human blood lipids: saturation at the pathway level, intact resolution at the gene level

The GLGC lipid meta-analyses ($n \approx 1.3$ million) are the opposite regime from the maize analysis: very large samples, well-characterised traits, and gene-level signal everywhere. We analysed LDL-C and HDL-C through an identical pipeline—same reference panel, same gene annotation, same Reactome sets, same window, same number of permutations—so that the only thing differing between them is the phenotype. Together they produce one clearly negative result and one clearly positive one, and the pair says more than either alone.

3.9.1 The self-contained null is false almost everywhere

At the gene level, 44.1% of tested genes had $p < 0.05$ for LDL-C and 11.2% reached genome-wide significance; for HDL-C the figures were 52.7% and 14.8%, with a median gene-level p-value of 0.038. The consequence for self-contained pathway testing is arithmetical rather than statistical. A *randomly chosen* set of fifty genes contains on average 5.6 genome-wide significant genes under LDL-C and 7.4 under HDL-C, and contains at least one with probability above 0.99 in both. The self-contained null hypothesis—that the genes in a set carry no more association than the background—is therefore false for essentially any set of genes one could assemble, including sets with no biological coherence whatever.

The output reflects this. Of the 1,390 Reactome pathways, 1,366 were significant under Westfall–Young family-wise control at $\alpha = 0.05$ for LDL-C and 1,384 for HDL-C. The natural first explanation is an inadequate multiple-testing correction, and we can rule it out: the Westfall–Young procedure controls the family-wise error rate exactly under the realized between-pathway dependence, and it removed essentially nothing. The rejections are not false positives in the sense a correction addresses. The null really is false for those sets. The problem is the hypothesis, not the threshold.

A second, subtler consequence is that the ranking acquires a strong dependence on pathway size. The twenty highest-ranked pathways had a median of 159 genes for LDL-C and 182 for HDL-C, against a median of 29 across all 1,390 sets, and the correlation between $\log_{10}$ set size and the pathway statistic was -0.825. In a saturated genome a large set samples more of an enriched background, so it accumulates more evidence for a reason that has nothing to do with the trait. Any interpretation of the ordering has to be made against that gradient, and we return to it below.

3.9.2 Two traits, different genes, the same pathway

LDL-C and HDL-C are biologically distinct arms of lipoprotein metabolism, and their gene-level results differ accordingly (Figure 10A). Across the 18,563 genes analysed in both, the Spearman correlation of gene-level Z statistics was 0.393. Of the top hundred genes in each trait, 13 were shared, and of the genes reaching genome-wide significance in either, the Jaccard overlap was 0.208 (830 shared of 3,987). These are not two readings of the same signal.

Despite that, the highest-ranked pathway of the 1,390 tested was the same in both: `REACTOME_PLASMA_LIPOPROTEIN_ASSEMBLY_REMODELING_AND_CLE`. At 74 genes it is well below the 159–182 median size of the surrounding top twenty, so it reaches first place against the size gradient of Section 3.9.1 rather than because of it.

The reason the two traits converge is visible in which genes carry the pathway in each (Figure 10B). Among its 74 members, the strongest LDL-C contributions come from the receptor-uptake arm—*LDLR* ($p = 3.1 \times 10^{-24}$), *PCSK9* (7.8×10^{-23}), *LDLRAP1* (7.4×10^{-22}), with *APOB*, *APOE* and *MTTP* close behind—while for HDL-C they come from the reverse-transport and lipolysis arm: *ABCA1* (1.3×10^{-33}), *LIPC* (5.8×10^{-29}), *SCARB1* (1.9×10^{-22}), *LPL* (8.9×10^{-22}), with *APOA1*, *ANGPTL3* and *LMF1* following. Only five of the fifteen leading drivers are common to both traits.

This is the behaviour a pathway method exists to produce. Two GWAS that disagree about which genes matter are nonetheless aggregated onto the same biological process, because that process contains both arms. The convergence is not a restatement of a shared gene-level signal; the gene-level signals are largely disjoint.

Two qualifications belong with the claim. First, the two traits are not independent replicates at the pathway level: across all 1,390 sets the Spearman correlation of the pathway statistic was 0.863, and 75 of the top hundred pathways were shared (Figure 10C). That correlation is driven by the size-dependent background of Section 3.9.1, which is common to both traits, rather than by the

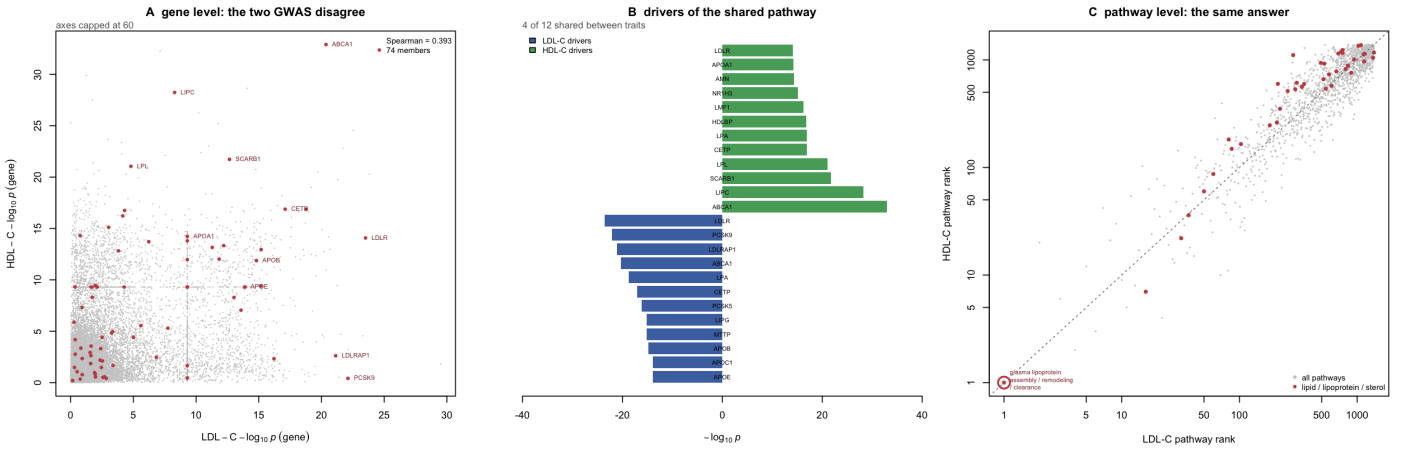


Figure 10 LDL-C and HDL-C reach the same pathway from different genes. **A** Gene-level $-\log_{10} p$ for the 18,563 genes analysed in both traits; the 74 members of the top-ranked pathway are highlighted and selected drivers labelled. The two GWAS agree only moderately (Spearman 0.393; 13 of the top hundred genes shared). **B** Within that pathway, the twelve strongest contributors in each trait. LDL-C is carried by the receptor-uptake arm and HDL-C by the reverse-transport and lipolysis arm, with five of fifteen leading drivers in common. **C** Pathway ranks in the two traits, lipid-related sets highlighted and the shared top pathway circled; it ranks first in both. The broad agreement away from the top reflects the size-dependent background of Section 3.9.1 rather than shared leading signal.

leading hits—which is precisely why the driver-level comparison, not the aggregate correlation, is the evidence for the claim above. Second, the phenotypes themselves are genetically correlated to a degree we have not estimated here; the gene-level Spearman of 0.393 is the quantity that actually enters CATFISH and is reported in its place.

Below the first rank, large generic sets—sumoylation, Golgi traffic, cellular response to chemical stress, rRNA processing—are interleaved with the lipid pathways in both traits, again consistent with the size gradient. Lipid, lipoprotein and sterol sets nonetheless concentrate near the top of both orderings, and they move between traits in a biologically sensible way: PPARα-regulated lipid metabolism ranks 16th for LDL-C and 7th for HDL-C, while sphingolipid metabolism ranks 81st and 182nd respectively.

3.9.3 The adaptive TFisher component is defensible but not load-bearing

The analyses above use all five component tests. Because the soft TFisher component is adaptive over a grid of truncation thresholds and is by some margin the slowest to evaluate, we also ran the complete HDL-C analysis with it removed, at matched $B = 10{,}000$, as a direct ablation.

The two orderings are almost identical: the Spearman rank correlation between the four- and five-component results was 0.9994, the median rank change was three places, and the largest change over all 1,390 pathways was 104 places, with 44 sets moving more than fifty. Every lipid pathway moved slightly upward when TFisher was included—plasma lipoprotein assembly $2 \to 1$, PPARα $8 \to 7$, lipoprotein clearance $98 \to 87$. This matches the leave-one-out simulation of Section 3.4, where removing TFisher left the family-wise error rate unchanged at 0.042.

We report this because the adaptive grid is the component most open to the objection that taking a minimum across five thresholds without correcting for the search is anti-conservative. That objection is correct for the analytic component—its uncalibrated type I error reaches 0.1214 at strong LD—but the resampling applies the same grid search to every null draw, so the inflation cancels, and the calibrated component is the best behaved of the five (0.0461). The ablation confirms that nothing in the real-data conclusions depends on it either way.

3.9.4 Gene prioritization is unaffected

The three-layer score of Section 2.5 was computed over all 18,563 genes. Table 9 gives the top five and Supplementary Table 11 the top fifty.

LDLR ranked 1 and *PCSK9* 3—the receptor that clears LDL from plasma and the protease that degrades it, the two genes around which LDL-lowering pharmacology is built. *APOE* and *APOC1*, at the chromosome 19 apolipoprotein cluster, follow close behind. Three of the top five are therefore genes with direct causal roles in LDL-C biology.

The other two require care, because they illustrate what a gene-based test can and cannot do. *PVRL2* (chr19:45.35–45.39 Mb) lies 17 kb from *APOE*, and *SMARCA4* (chr19:11.07–11.17 Mb) lies 27 kb from *LDLR*; *KIAA1324* (chr1:109.66–109.75 Mb), just outside the top five, sits in the *SORT1/CELSR2/PSRC1* region. None of these is a lipid gene. Each is an LD neighbour of one, and ranks highly because the association signal at a locus is shared across the genes within it. This is the expected behaviour of every gene-based test, CATFISH included: the method ranks *loci*, and resolving the causal gene inside a locus requires evidence it does not use. We state this explicitly because a table of top-ranked genes invites the opposite reading.

Figure 11 shows where each of the top twenty genes sits in each layer, which separates genes carried by a single layer from those with concordant support across all three.

As a pre-specified check, all thirteen established LDL-C genes named in Section 2.7 were recovered, with a median rank of 20 out of 18,563 against a random expectation of 9,282, and 12 of 13 in the top 1%.

The informative detail is *which* layer produces this. The pathway ranks of the control genes are good but unremarkable—*LDLR* ranks 462nd of 9,727 pathway-annotated genes—while their GWAS and

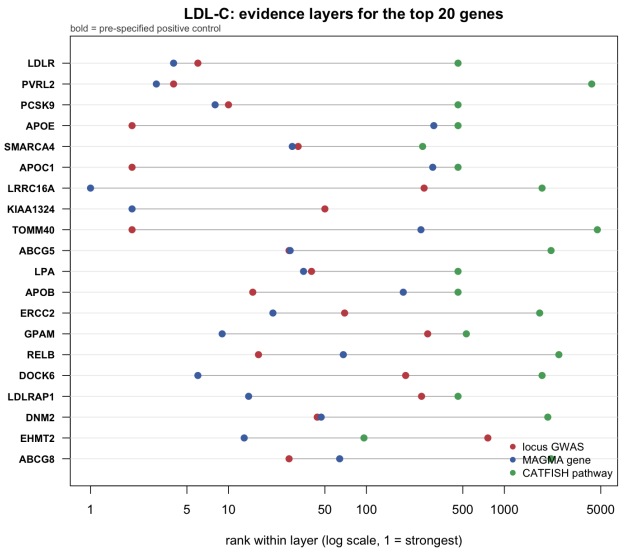


Figure 11 Evidence layers for the twenty highest-scoring genes in the LDL-C analysis. Each row is a gene; the three points give its rank within the locus-level GWAS layer, the MAGMA gene-level layer and the CATFISH pathway layer, on a log rank scale with one the strongest. Genes in bold are the pre-specified positive controls. The genetic layers place the controls near the top while their pathway ranks are unremarkable, which is the saturation of Section 3.9.1 seen gene by gene.

MAGMA ranks are near the top. The ordering is driven by the gene-level layers, and the pathway layer contributes little. That is the saturation result of Section 3.9.1 seen from the other side: saturation degrades the pathway layer specifically, because the self-contained hypothesis has stopped discriminating between gene sets, while leaving gene-level resolution intact. A composite score that weights pathway evidence at half the strength of direct genetic evidence degrades gracefully in exactly this regime, which was the reason for the weighting (Section 2.5.3) but not a situation it was designed against.

One implementation point affects this table directly and generalises beyond it. When the GWAS layer is ranked on probabilities reconstructed from GLGC and floored at 10^{-300}, 0 genes tie at the floor—including *LDLR*, *PCSK9*, *APOB*, *APOE*, *LPA* and *HMGCR*—and the layer cannot order the very genes it most needs to. Ranking on unfloored $-\log_{10} p$ removes the tie at no other cost. This is the same double-precision limit as Section 2.4, surfacing in a different part of the pipeline, and it is worth checking wherever a ranking is built from large-n GWAS summary statistics.

3.9.5 Signal density across datasets

Table 10 places the two real analyses in context. Gene-level signal density varies by more than an order of magnitude between the maize MLM analysis and the human meta-analysis, and the dependence of the pathway statistic on pathway size tracks it closely, from -0.179 in the sparsest dataset to -0.807 in the densest. This is not a bias to be corrected away. In a saturated regime larger sets genuinely do accumulate more association signal, because the background itself is enriched; the statistic reports something true about the data and uninteresting about the biology.

We note in passing that the maize GLM analysis, included here for contrast, shows a median gene-level p-value of 0.185, far below the 0.5 expected under the null. This is population-structure inflation rather than signal, and is why the MLM analysis is the one we report in Section 3.8. It also illustrates how readily a saturated-looking pathway result can arise from a mis-specified gene-level model rather than from biology, which is an argument for inspecting the gene-level null before interpreting any pathway output.

4 Discussion

4.1 What the Westfall–Young layer buys, and what it does not

The contribution of the Westfall–Young layer is narrow and should be stated narrowly. Reading the null matrix across its rows rather than down its columns costs nothing—the draws have already been taken—and it makes the family-wise statement exact under the actual overlap structure of the pathway database being used, without requiring that structure to be estimated or even described. It does not change any pathway's marginal evidence, which is what the column-wise reading gives, and it does not improve power. In our simulations it changed the significance threshold by a factor between 0.48 and 1.13 relative to Bonferroni, which is enough to matter for a borderline result and not enough to rescue an analysis that has gone wrong for other reasons.

The practical value is diagnostic as much as inferential. The ratio of the Bonferroni threshold to the realized Westfall–Young threshold is a single number that tells an analyst how far from independent their pathway family actually is. A ratio near one says the database behaves like a set of independent tests and Bonferroni is close to right; a ratio far from one says the correction in common use is substantially miscalibrated for this particular collection of gene sets. That number is cheap to compute as a by-product of the procedure and, as far as we are aware, is not routinely reported.

We were initially wrong about one aspect of this and correct the record here. We expected the size-standardized Westfall–Young variant of Section 2.3 to remove the dependence of pathway evidence on pathway size, on the reasoning that expressing each statistic against its own null bank puts all pathways on a common scale. It did not: in the maize data the size–evidence correlation moved from -0.807 to -0.800 in the human analysis and remained essentially unchanged elsewhere. The reason is visible once measured. The standard deviations of the per-size null banks differ by only a few percent across the pathway sizes we examined, so the standardization divides by a nearly constant factor and cannot remove a dependence that lives in the numerator. The size dependence is in the observed statistics, and in a saturated regime it is real accumulation of signal, not a scaling artefact. Standardization is still worth reporting because it changes individual rankings, but it is not a correction for size bias and should not be presented as one.

4.2 Saturation and the self-contained hypothesis

The LDL-cholesterol result in Section 3.9 is, we think, the most consequential finding in this paper, and it is not about CATFISH. Self-contained gene-set tests ask whether the genes in a set show more association than the null distribution of no association (Goeman and Bühlmann, 2007; Maciejewski, 2014). That question has a definite answer, and at $n \approx 1.3$ million for a highly polygenic trait the

answer is yes for almost every set of genes, because 11.2% of all genes are individually genome-wide significant. A random set of fifty genes rejects the self-contained null with probability 0.997. Nothing has malfunctioned; the hypothesis being tested has simply stopped discriminating.

Three consequences follow. First, as GWAS sample sizes grow, self-contained testing will fail on more traits, and it will fail silently, producing long lists of highly significant pathways that look like results. Second, no amount of multiple-testing sophistication fixes this: we applied exact family-wise control and 1,374 of 1,390 pathways remained significant. Reporting a more careful threshold on a saturated analysis produces a more carefully wrong answer. Third, the competitive hypothesis—whether the genes in a set are *more* associated than genes outside it—does not saturate in the same way, because the comparison is against the observed genome-wide background rather than against no association. For large, well-powered human GWAS, the competitive framing is the appropriate one, and the self-contained framing that CATFISH implements is best suited to the low-signal, strongly polygenic regime illustrated by the maize analysis, where no gene reaches genome-wide significance and a ranking is the only available output.

This is a boundary on the applicability of the method we are presenting, and we would rather state it than discover it in review. It also explains an initially puzzling observation: the pathways CATFISH ranked highest in the human analysis were things like spliceosome assembly and the TCA cycle, which have no particular relationship to lipid metabolism but are large and contain many of the genes that are significant in a saturated genome.

4.3 Numerical precision is a correctness issue

Section 2.4 reads like an implementation note and is not one. Had we not moved to log space, this paper would have reported a top-pathway list that was determined by file order, with every leading p-value equal to the double-precision floor, and nothing in the output would have indicated that anything was wrong. We encountered exactly this failure twice during development, once from ACAT overflow and once from underflow of the combined p-value, and in both cases the symptom was a plausible-looking table.

We therefore suggest that any pathway method intended for use on modern large-n GWAS summary statistics should state explicitly how it represents p-values below 10^{-300}, and that reviewers should ask. The arithmetic is not difficult; the problem is that the failure is invisible unless specifically checked for. A simple diagnostic is to count how many pathways share the smallest reported p-value. If that count exceeds one or two, the ranking below that point is not meaningful.

4.4 Practical recommendations

For users, our recommendations are as follows. Inspect the gene-level null before running any pathway analysis: a median gene-level p-value far from 0.5 indicates either genuine saturation or population-structure inflation, and these require different responses (the former argues for competitive testing, the latter for a mixed model). Calibrate against the correlated null whenever gene–gene correlations are available, and verify that the correlation file covers all chromosomes; in one of our own runs only chromosome 1 correlations had been parsed, and 85% of the genome was silently treated as independent, a failure that produced no error and no obviously wrong output. Report both the Westfall–Young family-wise p-values and the Bonferroni ones, along with their threshold ratio, since the comparison is informative about the database. Finally, in the low-signal regime, treat the ranking rather than the significance list as the deliverable, and validate it against independent biology where possible, as we have attempted to do with the maize analysis.

5 Limitations and future work

Several limitations bound what can be concluded from this work, and we list them explicitly.

No simulated GWAS. The simulations in Section 2.6 operate at the level of the omnibus statistic and its null, using an exchangeable correlation model for within-pathway dependence. We did not simulate genotypes, assign causal variants, generate phenotypes, and push them through the full GWAS–MAGMA–CATFISH pipeline. That experiment is the natural next step and is the one that would establish power against realistic genetic architectures rather than against latent mean shifts; it was deliberately excluded here rather than attempted partially. Consequently, the ranking results of Section 3.7 should be read as statements about the behaviour of the estimator, not as power curves.

Unreported correlations are set to zero. The correlation matrices of Section 2.3 are built by filling MAGMA's reported gene pairs into an identity matrix, so any pair MAGMA does not report is treated as independent. For distant gene pairs this is close to true and the matrix is banded by construction, but a nearby pair that goes unreported—because it fell outside a batch boundary, or because one gene was dropped—becomes a genuine understatement of dependence, in the anti-conservative direction. Section 3.6 quantifies the cost, and corrects an assumption we made before measuring it: the two readings do *not* inherit the limitation equally. The family-wise reading degrades about twice as fast as the marginal one, because the family minimum is a tail statistic and an error in the correlation structure does its worst damage in the tail. Verifying chromosome coverage of the correlation file catches the gross form of this failure; `10_pair_coverage.R` measures the fine-grained form, and we recommend reporting it alongside any family-wise p-value.

Exchangeable LD model. Real within-pathway gene–gene correlation is block-structured, driven by physical proximity, rather than exchangeable. Exchangeable R is a conservative stand-in for calibration questions, because it spreads a given average correlation evenly, but it does not reproduce the clustered dependence that arises when several pathway members sit in one LD block. The real-data analyses use empirical MAGMA correlations and are not subject to this limitation.

No competitive arm. We were unable to include a direct comparison against MAGMA's competitive gene-set test in this version, because the competitive entry point is not currently exported from the CATFISH package namespace. Given the argument of Section 4.2, that comparison is the single most important addition to make, and it is in progress.

Incomplete cross-species replication. The Westfall–Young analysis was run on maize and human data. The sorghum stem-volume dataset used in the original CATFISH paper could not be re-analysed under the new calibration because the MAGMA `.genes.raw` file, which the genome-wide correlated draw requires, was

not available in the archived results; its signal-density summary in Table 10 is reported from the gene-level output alone.

Tail resolution. Monte Carlo calibration resolves p-values only to $1/(B+1)$. CATFISH offers a generalized Pareto tail extrapolation for the extreme tail (Knijnenburg et al., 2009), but it requires a minimum number of valid null draws, and a single missing draw can silently disable it; in one of our human runs this caused 638 of 1,348 pathways to tie at the Monte Carlo floor. The threshold is now set to half the number of permutations rather than to the full number, but we have not yet verified the fix at production scale.

Dependence on upstream choices. Results remain sensitive to the SNP-to-gene window, the reference panel used for LD, gene model definitions, and the pathway database, as is true of every method in this class. The 10 kb windows used here are conventional but not optimal for any particular trait.

6 Conclusion

Pathway analysis of GWAS summary statistics faces two distinct dependence problems, and the literature has concentrated on the easier one. Correlation among genes within a pathway is widely acknowledged and routinely modelled; correlation among pathways, induced by the shared gene membership that is intrinsic to how pathway databases are built, is usually handled with a correction that assumes it away. We have added a Westfall–Young max-statistic layer to CATFISH that controls the family-wise error rate exactly under that dependence, requires nothing to be estimated about it, and yields as a by-product a direct measure of how far Bonferroni is from correct for any given database.

Two further findings matter beyond this particular method. Computing the omnibus in log space is a correctness requirement at modern GWAS sample sizes, not an optimisation, and the failure it prevents is silent. And self-contained pathway testing saturates: in a GWAS of 1.3 million individuals, 11.2% of genes are genome-wide significant, a random fifty-gene set rejects the self-contained null with probability 0.997, and exact family-wise error control leaves 1,374 of 1,390 pathways significant. That is a property of the hypothesis, not of the correction. CATFISH, and self-contained pathway testing generally, is best suited to the low-signal polygenic regime, where the maize grain-nitrogen analysis recovered L-homoserine biosynthesis as the leading pathway and a known aspartate kinase isozyme as the leading gene out of 34,873, in agreement with independent prior work.

7 Conflicts of interest

The authors declare that they have no competing interests.

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9 Data availability

The GLGC LDL-cholesterol summary statistics are available from the Global Lipids Genetics Consortium. The CATFISH R package and all code used to generate the figures, tables and simulations underlying this article are available at github.com/nirwantandukar/CATFISH, in the `catfish2/` directory.

10 Author contributions statement

N.T. conceived and implemented the method, designed and ran the simulations, analysed the real data, and wrote the manuscript. J.-Y.T. advised on the statistical framework and calibration. R.R.-A. supervised the work and revised the manuscript. All authors read and approved the final manuscript.

11 Acknowledgments

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Table 4 Component-breakage stress test on overlapping pathways. Each of k component tests is miscalibrated by raising its p-value to the power γ; $\gamma < 1$ makes a component conservative and $\gamma > 1$ anti-conservative. Marginal columns are per-pathway type I error; FWER columns are family-wise error rates over the K pathways tested, measured on held-out draws. Nominal level $\alpha = 0.05$.

LD	ρ	m	Direction	γ	k	Marginal		FWER	
						Analytic	MVN	WY	Bonf.
independent	0.0	20	anticonservative	1.4	0	0.0405	0.0508	0.0620	0.0240
independent	0.0	20	anticonservative	1.4	1	0.0666	0.0515	0.0540	0.1153
independent	0.0	20	anticonservative	1.4	2	0.0883	0.0518	0.0527	0.1347
independent	0.0	20	anticonservative	1.4	3	0.1012	0.0517	0.0607	0.2080
independent	0.0	20	anticonservative	1.4	4	0.1132	0.0513	0.0613	0.2407
independent	0.0	50	anticonservative	1.4	0	0.0404	0.0505	0.0600	0.0240
independent	0.0	50	anticonservative	1.4	1	0.0663	0.0496	0.0533	0.0927
independent	0.0	50	anticonservative	1.4	2	0.0879	0.0510	0.0547	0.1033
independent	0.0	50	anticonservative	1.4	3	0.1082	0.0503	0.0573	0.1553
independent	0.0	50	anticonservative	1.4	4	0.1273	0.0504	0.0587	0.1867
moderate	0.3	20	anticonservative	1.4	0	0.0616	0.0475	0.0460	0.1180
moderate	0.3	20	anticonservative	1.4	1	0.0925	0.0480	0.0480	0.2360
moderate	0.3	20	anticonservative	1.4	2	0.1118	0.0485	0.0500	0.2520
moderate	0.3	20	anticonservative	1.4	3	0.1259	0.0484	0.0460	0.3180
moderate	0.3	20	anticonservative	1.4	4	0.1387	0.0481	0.0473	0.3480
moderate	0.3	50	anticonservative	1.4	0	0.1024	0.0491	0.0520	0.1880
moderate	0.3	50	anticonservative	1.4	1	0.1365	0.0479	0.0527	0.2747
moderate	0.3	50	anticonservative	1.4	2	0.1542	0.0488	0.0527	0.2793
moderate	0.3	50	anticonservative	1.4	3	0.1806	0.0493	0.0560	0.3313
moderate	0.3	50	anticonservative	1.4	4	0.2061	0.0490	0.0540	0.3600
strong	0.6	20	anticonservative	1.4	0	0.0910	0.0530	0.0660	0.1640
strong	0.6	20	anticonservative	1.4	1	0.1258	0.0530	0.0653	0.2667
strong	0.6	20	anticonservative	1.4	2	0.1429	0.0529	0.0660	0.2707
strong	0.6	20	anticonservative	1.4	3	0.1587	0.0531	0.0640	0.3247
strong	0.6	20	anticonservative	1.4	4	0.1730	0.0530	0.0660	0.3473
strong	0.6	50	anticonservative	1.4	0	0.1371	0.0342	0.0420	0.2560
strong	0.6	50	anticonservative	1.4	1	0.1717	0.0349	0.0420	0.3273
strong	0.6	50	anticonservative	1.4	2	0.1850	0.0352	0.0420	0.3287
strong	0.6	50	anticonservative	1.4	3	0.2167	0.0359	0.0420	0.3660
strong	0.6	50	anticonservative	1.4	4	0.2483	0.0351	0.0420	0.3840
independent	0.0	20	conservative	0.7	0	0.0405	0.0508	0.0620	0.0240
independent	0.0	20	conservative	0.7	1	0.0320	0.0512	0.0600	0.0127
independent	0.0	20	conservative	0.7	2	0.0232	0.0508	0.0600	0.0120
independent	0.0	20	conservative	0.7	3	0.0163	0.0510	0.0493	0.0060
independent	0.0	20	conservative	0.7	4	0.0096	0.0510	0.0493	0.0040
independent	0.0	50	conservative	0.7	0	0.0404	0.0505	0.0600	0.0240
independent	0.0	50	conservative	0.7	1	0.0314	0.0506	0.0593	0.0153

Table 5 Leave-one-out component sensitivity on overlapping pathways. Each row removes a single component and rebuilds the null bank, the marginal calibration and the Westfall–Young threshold from scratch. Nominal level $\alpha = 0.05$.

LD	ρ	m	Dropped	Marginal		FWER	
				Analytic	MVN	WY	Bonf.
independent	0.0	20	none	0.0405	0.0508	0.0620	0.0240
independent	0.0	20	ACAT	0.0358	0.0514	0.0620	0.0300
independent	0.0	20	Fisher	0.0355	0.0512	0.0520	0.0100
independent	0.0	20	Stouffer	0.0355	0.0527	0.0560	0.0160
independent	0.0	20	minP	0.0368	0.0516	0.0640	0.0300
independent	0.0	20	TFisher	0.0584	0.0513	0.0620	0.0340
independent	0.0	50	none	0.0404	0.0505	0.0600	0.0240
independent	0.0	50	ACAT	0.0349	0.0495	0.0560	0.0260
independent	0.0	50	Fisher	0.0358	0.0506	0.0600	0.0100
independent	0.0	50	Stouffer	0.0369	0.0512	0.0620	0.0160
independent	0.0	50	minP	0.0388	0.0490	0.0560	0.0260
independent	0.0	50	TFisher	0.0573	0.0509	0.0520	0.0260
moderate	0.3	20	none	0.0616	0.0475	0.0460	0.1180
moderate	0.3	20	ACAT	0.0596	0.0468	0.0460	0.1240
moderate	0.3	20	Fisher	0.0525	0.0477	0.0560	0.0940
moderate	0.3	20	Stouffer	0.0486	0.0497	0.0460	0.0720
moderate	0.3	20	minP	0.0612	0.0467	0.0460	0.1240
moderate	0.3	20	TFisher	0.0817	0.0477	0.0460	0.1260
moderate	0.3	50	none	0.1024	0.0491	0.0520	0.1880
moderate	0.3	50	ACAT	0.1040	0.0481	0.0520	0.1960
moderate	0.3	50	Fisher	0.0902	0.0480	0.0480	0.1660
moderate	0.3	50	Stouffer	0.0782	0.0497	0.0520	0.1520
moderate	0.3	50	minP	0.1101	0.0479	0.0520	0.1960
moderate	0.3	50	TFisher	0.1217	0.0492	0.0520	0.1960
strong	0.6	20	none	0.0910	0.0530	0.0660	0.1640
strong	0.6	20	ACAT	0.0908	0.0527	0.0660	0.1740
strong	0.6	20	Fisher	0.0779	0.0530	0.0680	0.1360
strong	0.6	20	Stouffer	0.0679	0.0539	0.0620	0.1200
strong	0.6	20	minP	0.0931	0.0527	0.0660	0.1740
strong	0.6	20	TFisher	0.1123	0.0533	0.0660	0.1740

Table 6 Per-component calibration. Each component test is calibrated against its own null rather than being combined into the omnibus, at $m = 50$. Analytic columns assume gene independence; calibrated columns use the correlated null. Nominal $\alpha = 0.05$.

LD	Component	Analytic	Calibrated	WY	λ_{an}	λ_{cal}
independent	ACAT	0.0526	0.0512	0.0460	0.992	0.986
independent	Fisher	0.0501	0.0506	0.0540	1.015	1.014
independent	Stouffer	0.0496	0.0495	0.0520	0.999	1.005
independent	minP	0.0523	0.0504	0.0460	0.988	0.977
independent	TFisher	0.0438	0.0476	0.0600	1.216	0.995
independent	Omnibus	0.0404	0.0505	0.0600	1.055	1.005
moderate	ACAT	0.0543	0.0540	0.0840	1.063	1.072
moderate	Fisher	0.1222	0.0480	0.0520	0.954	1.062
moderate	Stouffer	0.1401	0.0454	0.0480	1.064	1.062
moderate	minP	0.0490	0.0551	0.0860	0.885	1.081
moderate	TFisher	0.0751	0.0492	0.0380	1.378	0.918
moderate	Omnibus	0.1024	0.0491	0.0520	1.085	1.072
strong	ACAT	0.0547	0.0518	0.0660	0.980	0.968
strong	Fisher	0.1566	0.0372	0.0420	0.789	0.941
strong	Stouffer	0.1805	0.0323	0.0400	0.991	0.941
strong	minP	0.0452	0.0537	0.0600	0.508	0.995
strong	TFisher	0.1214	0.0461	0.0400	1.981	1.045
strong	Omnibus	0.1371	0.0342	0.0420	0.997	0.959

Table 7 Robustness to an incomplete gene–gene correlation file. Data are generated from the true correlation structure; calibration is built from a matrix in which the stated fraction of off-diagonal entries has been zeroed, as an unreported gene pair would be. Marginal columns are per-pathway type I error, FWER columns are family-wise rates over K pathways. Nominal $\alpha = 0.05$.

LD	Dropped	Marginal		FWER		λ	
		Analytic	MVN	WY	Bonf.	Analytic	MVN
moderate	0%	0.0987	0.0501	0.0625	0.1850	1.096	1.090
moderate	25%	0.0987	0.0622	0.0775	0.1850	1.096	1.041
moderate	50%	0.0987	0.0877	0.1475	0.1850	1.096	1.128
moderate	75%	0.0987	0.0901	0.1700	0.1850	1.096	1.053
moderate	100%	0.0987	0.1168	0.2200	0.1850	1.096	1.029
strong	0%	0.1458	0.0654	0.0625	0.2475	1.116	1.029
strong	25%	0.1458	0.0957	0.1250	0.2475	1.116	1.128
strong	50%	0.1458	0.1149	0.1725	0.2475	1.116	1.018
strong	75%	0.1458	0.1257	0.2100	0.2475	1.116	0.926
strong	100%	0.1458	0.1658	0.2825	0.2475	1.116	1.141

Table 8 Ranking of truly enriched pathways and family-wise significance thresholds. Each replicate contains N_{P} pathways of which N_{true} carry a latent mean shift μ. Median rank is the median within-replicate rank of the enriched pathways (1 = best); random expectation is $N_{\mathrm{P}}/2$. The WY threshold is the 5th percentile of the minimum null omnibus p-value across the family; the ratio column is Bonferroni/WY.

LD	ρ	m	Rank (analytic)	Rank (MVN)	Random	WY thresh.	Bonf./WY
independent	0.0	20	8.0	7.9	150	1.97×10^{-4}	0.84
independent	0.0	50	8.0	8.0	150	3.00×10^{-4}	0.56
moderate	0.3	20	10.0	10.0	150	3.50×10^{-4}	0.48
moderate	0.3	50	9.7	9.7	150	3.50×10^{-4}	0.48
strong	0.6	20	17.7	17.7	150	1.47×10^{-4}	1.13
strong	0.6	50	17.6	17.6	150	3.00×10^{-4}	0.56

Table 9 Top five prioritized genes for LDL cholesterol. Rank columns are the gene's position within each evidence layer: locus-level GWAS (18563 genes with an overlapping SNP), MAGMA gene-level association (18563 genes), and CATFISH pathway enrichment (9727 pathway-annotated genes). Score is the composite of Section 2.5. The full top fifty are in Supplementary Table S1.

Rank	Gene	Score	GWAS	MAGMA	Pathway	Leading pathway
1	*LDLR*	8.11	6	4	462	plasma lipoprotein assembly remod…
2	*PVRL2*	7.73	4	3	4298	immunoregulatory interactions bet…
3	*PCSK9*	7.53	10	8	462	plasma lipoprotein assembly remod…
4	*APOE*	6.80	2	308	462	plasma lipoprotein assembly remod…
5	*SMARCA4*	6.76	32	29	256	atp dependent chromatin remodelers

Table 10 Gene-level signal density and pathway-size dependence across the four GWAS analysed. The last column is the Spearman correlation between pathway size and the pathway-level evidence ($-\log_{10}$ omnibus p); more negative values indicate that larger pathways receive systematically stronger evidence. n.c., not computed.

Dataset	Genes GW significant	Median gene p	Size correlation
Maize grain N (MLM)	0.00%	0.468	-0.179
Sorghum stem volume	0.86%	0.440	n.c.
Maize grain N (GLM)	1.15%	0.185	-0.667
Human LDL-C (GLGC, EUR)	11.2%	0.083	-0.807

Supplementary material

Table 11 Top fifty prioritized genes for LDL cholesterol, with per-layer ranks. Layer sizes as in Table 9 (18563 / 18563 / 9727 genes).

Rank	Gene	Score	GWAS	MAGMA	Pathway	Leading pathway
1	*LDLR*	8.11	6	4	462	plasma lipoprotein assembly remod...
2	*PVRL2*	7.73	4	3	4298	immunoregulatory interactions bet...
3	*PCSK9*	7.53	10	8	462	plasma lipoprotein assembly remod...
4	*APOE*	6.80	2	308	462	plasma lipoprotein assembly remod...
5	*SMARCA4*	6.76	32	29	256	atp dependent chromatin remodelers
6	*APOC1*	6.65	2	302	462	plasma lipoprotein assembly remod...
7	*LRRC16A*	6.57	262	1	1876	factors involved in megakaryocyte...
8	*KIAA1324*	6.54	50	2	–	–
9	*TOMM40*	6.25	2	248	4722	protein localization
10	*ABCG5*	6.23	28	28	2180	disorders of transmembrane transp...
11	*LPA*	6.20	40	35	462	plasma lipoprotein assembly remod...
12	*APOB*	6.16	15	185	462	plasma lipoprotein assembly remod...
13	*ERCC2*	6.14	70	21	1803	rna polymerase i transcription
14	*GPAM*	6.08	278	9	530	estrogen dependent gene expression
15	*RELB*	6.04	16	68	2484	somatic hypermutation of immunogl...
16	*DOCK6*	6.01	192	6	1876	factors involved in megakaryocyte...
17	*LDLRAP1*	5.97	251	14	462	plasma lipoprotein assembly remod...
18	*DNM2*	5.93	44	47	2065	clathrin mediated endocytosis
19	*EHMT2*	5.89	759	13	96	cellular senescence
20	*ABCG8*	5.87	28	64	2180	disorders of transmembrane transp...
21	*SPC24*	5.85	33	39	4078	mitotic prometaphase
22	*IGF2R*	5.78	138	16	2065	clathrin mediated endocytosis
23	*ABCA1*	5.78	273	18	462	plasma lipoprotein assembly remod...
24	*MAP2K6*	5.75	119	128	96	cellular senescence
25	*ANKDD1B*	5.70	36	19	–	–
26	*APOC2*	5.68	12	626	462	plasma lipoprotein assembly remod...
27	*CETP*	5.58	120	59	462	plasma lipoprotein assembly remod...
28	*HLA-DQA1*	5.57	223	26	862	interferon gamma signaling
29	*CBLC*	5.56	9	106	–	–
30	*BSND*	5.44	63	37	5394	ion channel transport
31	*GATAD2A*	5.41	53	389	256	atp dependent chromatin remodelers
32	*HIST1H2AC*	5.40	184	271	96	cellular senescence
33	*CEACAM16*	5.39	18	76	–	–
34	*PPP1R13L*	5.37	70	48	3710	regulation of tp53 activity
35	*HMGCR*	5.35	22	356	1304	metabolism of steroids
36	*USP24*	5.35	20	165	3206	ub specific processing proteases
37	*CELSR2*	5.33	8	213	–	–
38	*KANK2*	5.31	31	54	–	–
39	*HIST1H4C*	5.31	174	365	96	cellular senescence
40	*POLK*	5.31	35	195	2780	homology directed repair
41	*APOC4*	5.30	12	1174	462	plasma lipoprotein assembly remod...
42	*PHC1*	5.30	549	72	96	cellular senescence
43	*NPC1L1*	5.26	127	20	8228	digestion and absorption
44	*BCAM*	5.26	5	380	–	–
45	*GSTM2*	5.20	252	30	1581	phase ii conjugation of compounds

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